

Quantiles over data streams: experimental comparisons, new analyses, and further improvements

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Abstract A fundamental problem in data management and analysis is to generate descriptions of the distribution of data. It is most common to give such descriptions in terms of the cumulative distribution, which is characterized by the quantiles of the data. The design and engineering of efficient methods to find these quantiles has attracted much study, especially in the case where the data are given incrementally, and we must compute the quantiles in an online, streaming fashion. While such algorithms have proved to be extremely useful in practice, there has been limited formal comparison of the competing methods, and no comprehensive study of their performance. In this paper, we remedy this deficit by providing a taxonomy of different methods and describe efficient implementations. In doing so, we propose new variants that have not been studied before, yet which outperform existing methods. To illustrate this, we provide detailed experimental comparisons demonstrating the trade-offs between space, time, and accuracy for quantile computation.

Keywords Data stream algorithms · Quantiles · Ordinary least squares

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1 Introduction

Given a large amount of data, a first and foundational problem is to describe the data distribution. If the data follow a known distribution family, such as Gaussian, they can be described succinctly by the parameters of the distribution. This is rarely the case in practice, which thus calls for nonparametric methods. Quantiles are the most commonly used nonparametric representation for data distributions. They correspond to the cumulative distribution function (cdf), which in turn yields the probability distribution function (pdf). Thus, quantile computation is arguably one of the most fundamental problems in data analysis. For example, rankings are often expressed in terms of percentiles, such as for giving results of standardized testing, or measuring children's physical development. Distributions are commonly compared via quantiles, in the form of quantile–quantile plots, which leads to the *Kolmogorov–Smirnov divergence*, one of the most commonly used distance measures between distributions.

Computing the quantiles has significant practical importance: Standard statistical packages, such as R and Excel, include functions to compute the median and other quantiles. In the Sawzall language that is the basis for all of Google's log data analysis, quantile is one of the seven basic operators defined (the others include sum, max, top- k , and count-distinct) [24]. The quantiles also play an important role in network health monitoring for Internet service providers [8] and data collection in wireless sensor networks [26].

The problem is also intellectually interesting enough to have attracted a lot of prior study, from both the algorithms and the database community, sometimes investigated under the name of “the selection problem” or “order statistics.” Algorithmic interest dates back to at least 1973, when the celebrated *linear-time selection* algorithm was invented [4]. In the past 35 years, this problem has received particular atten-

tion in the streaming model, i.e., the data elements arrive one by one in a streaming fashion, and the algorithm only has limited memory to work with. There have been numerous algorithms proposed in this setting, using a variety of different techniques and offering different performance guarantees [7, 12, 13, 15, 18, 21–23, 27]. In addition, there have been many studies on variations and extensions of the problem, such as computing quantiles over sliding windows [3], over distributed data [1, 16, 17, 26], continuous monitoring of quantiles [9, 30], biased quantiles [10], computing quantiles using GPUs [14].

The median has long been recognized as a more stable statistic of data distribution than, say, the average, in the sense that it is very robust to outliers. The quantiles are a natural generalization of the median. Let S be a (multi)set of n elements drawn from a totally ordered universe. Recall that the ϕ -quantile of S , for some $0 < \phi < 1$, is the element whose rank is $\lfloor \phi n \rfloor$ in S , where the rank of an element x is the number of elements in S smaller than x .

The quantiles can be easily found by sorting if sufficient space is available. The problem becomes significantly more challenging in the streaming model, which is the focus of this work. It dates back to 1980, when Munro and Paterson [23] showed that any algorithm that computes the median exactly with p passes over the data has to use $\Omega(n^{1/p})$ space. Thus, approximation is necessary for any streaming quantile algorithm using sublinear space. Recall that a streaming algorithm is one that makes one pass over the data and performs the desired computation. Often, the algorithm is not given the knowledge of n , the length of the stream, so that the algorithm has to be ready to stop and provide the results at any time. This corresponds to the practical setting where the stream is conceptually an infinite sequence of elements, and the algorithm should always be ready to provide the results for the data seen so far. In line with most prior work, we also adopt this requirement.

Subsequently, the problem of computing approximate quantiles over streaming data has been widely studied in the past three decades (which will be reviewed shortly). The commonly used notion of approximation for this problem is the following: for an error parameter $0 < \varepsilon < 1$, the ε -approximate ϕ -quantile is any element with rank between $(\phi - \varepsilon)n$ and $(\phi + \varepsilon)n$. Since quantiles are used for approximating the data distribution anyway, and the input data are often noisy in itself, allowing some errors in the computed quantiles is often tolerable.

However, despite the importance of the problem and the many efforts devoted, a complete and clear picture of the problem still appears elusive, both theoretically and empirically. We lack matching upper and lower bounds for the problem, which constitutes a top open problem in data stream algorithms (see <http://sublinear.info/2>). Moreover, existing empirical studies are both incomplete and outdated. In this

work, we set out to address this issue, and carry out an extensive experimental comparison of various quantile algorithms that have not been compared before. In doing so, we also propose new variants that have not been studied before, yet which turn out to perform the best.

1.1 Classification of algorithms

Depending on different models, algorithms for computing quantiles of data streams can be classified along the following axes:

1. Whether elements can only be added or can be both inserted and deleted.

In the *cash register* model, elements arrive one by one in the stream and they are never removed. In the *turnstile model*, the stream consists of a sequence of updates where each update either inserts an element or deletes one, but a deletion cannot delete an element that does not exist. When there are duplicates, this means that the multiplicity of any element cannot go negative.

2. What operations are allowed on the elements.

In the *comparison model*, the algorithm can only access the elements through comparisons. Implicitly, this means that the algorithm must store a set of elements that it has observed from the stream (together with some extra information), and only return from this set as quantiles in the end, namely it cannot “create” or “compute” elements to return. In the *fixed-universe model*, the elements are integers in the universe $[u] = \{0, \dots, u - 1\}$. Here, the algorithm is allowed to perform bit manipulation tricks and return elements that may have never appeared in the stream as quantiles provided they satisfy the approximation guarantees. Clearly, the comparison model is more restrictive, so any comparison-based algorithm also works in the fixed-universe model, but not vice versa. However, the benefit of comparison-based algorithms is that they can handle elements that cannot be easily mapped to a fixed universe $[u]$, such as variable-length strings or user-defined types.¹

3. Whether the algorithm is deterministic or randomized.

We are not aware of any Las Vegas quantile algorithms, so we will only consider Monte Carlo randomization, where an algorithm may return an incorrect quantile (i.e., exceeding the stated ε error) with a small probability. We usually consider the probability that the algorithm returns *all* quantiles correctly, but this will be the case

¹ Note that floating-point numbers in standard representations (e.g., IEEE 754) can be mapped to integers in a fixed universe in an order-preserving fashion.

as long as it is correct on the $1/\varepsilon - 1$ quantiles for $\phi = \varepsilon, 2\varepsilon, \dots, 1 - \varepsilon$. The quantiles in between any two of these quantiles will thus have error at most 2ε , and scaling ε down by a factor of 2 will restore the ε -approximation guarantee for all quantiles. To simplify the bounds, most theoretical analyses make this probability a constant. This probability can always be boosted using standard techniques; in practice, due to the looseness of the analysis, it suffices to set the success probability to a reasonable constant.

1.2 Existing quantile algorithms and our new findings

1.2.1 The cash register model

In their pioneering paper [23], Munro and Paterson also gave a p -pass algorithm for computing exact quantiles. Although not analyzed explicitly, the first pass of the algorithm yields a streaming algorithm for computing ε -approximate quantiles using $O(\frac{1}{\varepsilon} \log^2(\varepsilon n))$ space. This fact was made more explicit by Manku et al. [21], who also proposed another algorithm that is empirically better, though it has the same worst-case space bound. In 2001, Greenwald and Khanna [15] designed a quite ingenious algorithm (referred to as the GK algorithm below) and showed that it uses $O(\frac{1}{\varepsilon} \log(\varepsilon n))$ space in the worst case. But interestingly, their experimental study implements a simplified algorithm (referred to as **GKAdaptive** below), for which it is not clear if the $O(\frac{1}{\varepsilon} \log(\varepsilon n))$ space bound still holds. Nevertheless, they showed that this algorithm empirically outperforms that of Manku et al. [21]. All these algorithms are deterministic and comparison-based. Hung and Ting [18] showed an $\Omega(\frac{1}{\varepsilon} \log \frac{1}{\varepsilon})$ space lower bound for such algorithms. In this category, the GK algorithm is generally considered to be the best, both theoretically and empirically (in its respective versions).

In 2004, Shrivastava et al. [26] designed a deterministic, fixed-universe algorithm, called q -digest, that uses $O(\frac{1}{\varepsilon} \log u)$ space. This algorithm was designed for quantile computation in sensor networks and is a *mergeable summary* [1], a model that is more general than streaming. But no better fixed-universe algorithm is known in the streaming model. Note that the $\log u$ and $\log(\varepsilon n)$ terms are not comparable in theory, and [26] did not include an experimental comparison with the GK algorithm.

Randomized algorithms have also been investigated. Classic results [28] show that a random sample of size $O(\frac{1}{\varepsilon^2} \log \frac{1}{\varepsilon})$ preserves all quantiles within ε error with at least a constant probability. This fact was reproved in [21] and exploited for computing quantiles by feeding a random sample to a deterministic algorithm. But this algorithm requires the a priori knowledge of n , so it is not a true streaming algorithm. Later,

Manku et al. [22] proposed a randomized algorithm (henceforth referred to as **MRL99**) that does not need the knowledge of n and showed that its space requirement is $O(\frac{1}{\varepsilon} \log^2 \frac{1}{\varepsilon})$. Note that the $\log^2 \frac{1}{\varepsilon}$ factor could be either larger or smaller than the $\log(\varepsilon n)$ factor of GK, and these two algorithms have not been compared experimentally. Subsequently, Agarwal et al. [1] gave a more complicated algorithm with a space complexity of $O(\frac{1}{\varepsilon} \log^{1.5}(\frac{1}{\varepsilon}))$ without implementation. Very recently, Felber and Ostrowsky [11] provided a randomized algorithm (also without implementation) for this problem that achieves $O(\frac{1}{\varepsilon} \log \frac{1}{\varepsilon})$ space cost. However, the hidden constant in the big-Oh is so substantially large that it makes this algorithm only of theoretical interest. Our prototype implementation of this algorithm confirmed that it is not competitive in practice with others, so we do not consider it further in this empirical study.

In this paper, we empirically compare **GKAdaptive**, q -digest, and **MRL99**. We omit results for the algorithms of Munro and Paterson [23] and the earlier algorithm of Manku et al. [21], since they have previously been demonstrated to be outperformed by the GK algorithm. We have also implemented **GKTheory** and found out that it does not perform as well as **GKAdaptive**, despite the $O(\frac{1}{\varepsilon} \log(\varepsilon n))$ space guarantee of the former.

Our experimental study reveals that **MRL99** generally performs the best, but it suffers from the following undesirable properties. First, it uses some fairly complex rules for maintaining its samples and sets its parameters delicately by solving an optimization problem, which increases implementation difficulty. Second, as the algorithm is difficult to analyze, the analysis given in [22] is quite pessimistic, resulting in an $O(\frac{1}{\varepsilon} \log^2(\frac{1}{\varepsilon}))$ bound. In practice, this means that for an error target ε , we often allocate more space than necessary. Through our experimental study, we observed that many of the details of **MRL99** were not actually needed, and the algorithm can be significantly simplified without affecting its performance. In addition, we give a new analysis on this simpler algorithm (referred to as **Random**), leading to an improved $O(\frac{1}{\varepsilon} \log^{1.5}(\frac{1}{\varepsilon}))$ bound. For a randomized algorithm, a subtle issue is whether the ε -guarantee holds for all quantiles or any one quantile. The latter case could be relevant when a specific quantile, e.g., the median, is needed. All the bounds stated above hold for the all-quantile guarantee; in fact, they cannot be improved even if the weaker, one-quantile guarantee is needed, except that for the **Random** algorithm, we show that the space can be further reduced to $O(\frac{1}{\varepsilon} \log \frac{1}{\varepsilon})$ for the one-quantile guarantee.

On the other hand, **GKAdaptive** remains the most competitive deterministic algorithm. However, the original paper [15] focused only on space usage and did not elaborate on the running time of the algorithm. In this paper, we have identified two different ways to implement the algorithm and investigated their practical performance.

Table 1 All algorithms evaluated in this paper

Algorithm	Space	Update time	Randomization	Model
GKAdaptive	–	$O(\log \text{Space})$	Deterministic	Comparison
GKTheory	$O(\frac{1}{\varepsilon} \log(\varepsilon n))$	$O(\log \frac{1}{\varepsilon} + \log \log(\varepsilon n))$	Deterministic	Comparison
FastQDigest	$O(\frac{1}{\varepsilon} \log u)$	$O(\log \frac{1}{\varepsilon} + \log \log u)$	Deterministic	Fixed universe
MRL99	$O(\frac{1}{\varepsilon} \log^2 \frac{1}{\varepsilon})$	$O(\log \frac{1}{\varepsilon})$	Randomized	Comparison
Random* (all quantiles)	$O(\frac{1}{\varepsilon} \log^{1.5} \frac{1}{\varepsilon})$	$O(\log \frac{1}{\varepsilon})$	Randomized	Comparison
Random* (one quantile)	$O(\frac{1}{\varepsilon} \log \frac{1}{\varepsilon})$	$O(\log \frac{1}{\varepsilon})$	Randomized	Comparison
Random subset sum	$O(\frac{1}{\varepsilon^2} \log^2 u \log(\frac{\log u}{\varepsilon}))$	$O(\frac{1}{\varepsilon^2} \log^2 u \log(\frac{\log u}{\varepsilon}))$	Randomized	Fixed universe
DCM	$O(\frac{1}{\varepsilon} \log^2 u \log(\frac{\log u}{\varepsilon}))$	$O(\log u \log(\frac{\log u}{\varepsilon}))$	Randomized	Fixed universe
DCS*	$O(\frac{1}{\varepsilon} \log^{1.5} u \log^{1.5}(\frac{\log u}{\varepsilon}))$	$O(\log u \log(\frac{\log u}{\varepsilon}))$	Randomized	Fixed universe

Those marked with * are new variants

1.2.2 The turnstile model

The turnstile model presents additional challenges, due to the deletions of elements. Attempts to adapt the above algorithms to this model can often be thwarted by finding particularly adversarial patterns of insertions and subsequent deletions. In fact, it can be argued that no comparison-based algorithm is possible using sublinear space under the turnstile model: Imagine that we first insert n elements and then delete all but one. Before the deletions, the algorithm has no information about which element will survive, and because the comparison-based model does not allow the creation or computation of elements to return, it has to retain all n elements. Therefore, all turnstile algorithms work only for a fixed universe, and are mostly randomized algorithms. Deterministic algorithms for the fixed-universe model have been provided: Ganguly and Majumder describe an algorithm which uses $O(\frac{1}{\varepsilon^2} \log^5 u \log(\frac{\log u}{\varepsilon}))$ space [12]. The high dependency on $\frac{1}{\varepsilon}$ and $\log u$ is not considered practical.

Existing algorithms in the turnstile model all make use of a *dyadic structure* imposed over the universe of possible elements. More precisely, we build $\log u$ levels, decomposing the universe $[u]$ as follows. In level 0, every integer in $[u]$ is by itself; in level i , the universe is partitioned into intervals of size 2^i ; the top level thus consists of only one interval $[0, u - 1]$. Every interval in every level in this hierarchy is called a *dyadic interval*. The algorithms make use of randomized *sketch* data structures which process a stream of updates in the turnstile model and allow the frequency of any element to be estimated [5, 7]. Each level keeps a frequency estimation sketch that can be used to estimate the total number of elements in any interval on that level. To find the rank of a given element x , we decompose the interval $[0, x - 1]$ into the disjoint union of at most $\log u$ dyadic intervals, one from each level. From the frequency estimation sketch, we estimate the number of elements in each dyadic interval, and then add them up. Then for any given ϕ , we can find an

approximate ϕ -quantile by doing a binary search on $[u]$ to find the largest element whose rank is below ϕn .

Different frequency estimation sketches have been proposed to instantiate this outline. Gilbert et al. [13] first proposed the *random subset sum* sketch for this purpose, which results in a size of $O(\frac{1}{\varepsilon^2} \log^2 u \log(\frac{\log u}{\varepsilon}))$. Later, Cormode and Muthukrishnan applied the Count-Min sketch in the dyadic structure (the resulting algorithm is referred to as DCM, for “Dyadic Count-Min”), reducing the overall size to $O(\frac{1}{\varepsilon} \log^2 u \log(\frac{\log u}{\varepsilon}))$ [7]. This remains the best bound in the turnstile model. In this paper, we propose to use the Count-Sketch [5] (the algorithm is thus referred to as DCS, for “Dyadic Count-Sketch”) and give a new analysis showing that it further reduces the space to $O(\frac{1}{\varepsilon} \log^{1.5} u \log^{1.5}(\frac{\log u}{\varepsilon}))$, which is the best bound for this problem under the turnstile model. We also made an experimental comparison of these different variants, which shows that DCS is not only theoretically the best, but also gives superior performance in practice.

Finally, to further improve the accuracy of DCS, we design a fast post-processing step to eliminate the discrepancies in the frequency estimates across different levels, using the *ordinary least squares* method. Experimental results show that this post-processing step can further reduce the error of DCS by 60–80 %.

Table 1 summarizes all the algorithms that we evaluate in this paper, in both the cash register and the turnstile model.

1.3 Relation to conference publication

This paper extends our earlier work [29], in which we identified GKAdaptive to be the best deterministic algorithm, Random to be the best randomized algorithm, while DCS the best algorithm in the turnstile model. In this paper, we develop several new ideas that lead to further improvements to these algorithms:

1. We give a new implementation of **GKAdaptive**, called **GKArray**, which uses buffering techniques to significantly improve the running time (Sect. 2.1.2).
2. We show how the new **Random** algorithm can be applied to the sliding window quantile algorithm of Arasu and Manku [3], leading to an improved bound of $O(\frac{1}{\varepsilon} \log^{1.5} \frac{1}{\varepsilon})$ for the one-quantile guarantee, which is the best theoretical result for this problem (Sect. 2.5). In practice, the new algorithm also significantly outperforms that of [3].
3. For **DCS**, we give a novel “post-processing” step to make better use of the estimates generated, to give substantially improved accuracy for this class of algorithms (Sect. 3.4). This post-processing step can be used not only as an offline improvement of the sketch at the end of the stream, but also in an online setting, leading to improved estimates at any point in time.

2 Cash register algorithms

In this section, we describe the cash register algorithms. Recall that in this model, there are only insertions in the stream. We use n to denote the current number of elements in the stream. We use $r(x)$ to denote the rank of x in all the elements received so far.

2.1 The GK algorithm

The GK algorithm [15] is a deterministic, comparison-based quantile algorithm. It maintains a list of tuples $L = \langle (v_i, g_i, \Delta_i) \rangle$, where the v_i 's are elements from the stream such that $v_i \leq v_{i+1}$. The g_i 's and Δ_i 's are integers satisfying the following conditions:

- (1) $\sum_{j \leq i} g_j \leq r(v_i) + 1 \leq \sum_{j \leq i} g_j + \Delta_i$;
- (2) $g_i + \Delta_i \leq \lfloor 2\varepsilon n \rfloor$.

Note that condition (1) gives both a lower and an upper bound on the possible ranks of v_i . Also, $g_i + \Delta_i - 1$ is the maximum possible number of elements between v_{i-1} and v_i , so (2) ensures that for any $0 < \phi < 1$, there must be an element in the list whose rank is within εn of ϕn . Thus, to extract the ϕ -quantile, we can find the smallest i such that $\sum_{j \leq i} g_j + \Delta_i > 1 + \lceil \phi N \rceil + \max_i (g_i + \Delta_i) / 2$ and then report v_{i-1} . It can be verified that this v_{i-1} will be a valid ε -approximate ϕ -quantile.

The list is initialized as $L = \langle (\infty, 1, 0) \rangle$. To insert a new element v , we find its successor in L , i.e., the smallest v_i such that $v_i > v$, and insert the tuple $(v, 1, \lfloor 2\varepsilon n \rfloor)$ right before v_i . We may also remove tuples: To remove (v_i, g_i, Δ_i) , we set $g_{i+1} \leftarrow g_i + g_{i+1}$ and remove the tuple from L . Note that

this may violate condition (2) for the next tuple, so we call a tuple *removable* if $g_i + g_{i+1} + \Delta_{i+1} \leq \lfloor 2\varepsilon n \rfloor$.

In order to keep $|L|$ small, the original paper [15] gave a fairly complex COMPRESS procedure to carefully select tuples to remove while maintaining (1). It is performed once every $\frac{1}{2\varepsilon}$ incoming elements. It has been shown that after the COMPRESS procedure, $|L|$ is at most $\frac{11}{2\varepsilon} \log(2\varepsilon n)$. COMPRESS can be done in time $O(|L|)$, so if it is performed only when $|L|$ doubles, its amortized cost is $O(1)$ per update. An insertion can be done in time $O(\log |L|)$ if we maintain a binary search tree on top of L ; therefore, the amortized per-element update time is $O(\log \frac{1}{\varepsilon} + \log \log(\varepsilon n))$.

2.1.1 Variant: GKAdaptive

The algorithm described was structured to permit theoretical analysis of the space cost; in the paper [15], the authors instead implemented the following variant:

1. To insert v , insert to L a tuple $(v, 1, g_i + \Delta_i - 1)$ instead of $(v, 1, \lfloor 2\varepsilon n \rfloor)$.
2. Following an insertion, try to find a removable tuple in L . If there is one, remove it; otherwise $|L|$ increases by 1.

Note that COMPRESS is never called in this variant, so the $O(\frac{1}{\varepsilon} \log(\varepsilon n))$ bound may not hold.

The original paper [15] did not specify how to find a removable tuple, as they did not focus on running time. There can be two ways to implement this efficiently. The first is to maintain a min-heap on the tuples in L ordered by $g_i + g_{i+1} + \Delta_{i+1}$. When a new tuple is inserted, we first check if the tuple itself is removable and remove it immediately if so. Otherwise, we check the top tuple in the heap and remove it if it is removable. If the top tuple in the heap is not removable, then no others are. When this happens, $|L|$ increases by 1. The heap can be maintained in $O(\log |L|)$ time per element, so the asymptotic update time is not affected. A quantile query can be answered by a binary search, which takes $O(\log |L|)$ time. We refer to this variant as **GKAdaptive**.

2.1.2 Variant: GKArray

A quite different way to implement the algorithm above is to defer some actions and operate in a “batch mode.” We store all tuples in L in an array instead of a list. We do not insert tuples into the array so we will not need the binary search tree. We remove the use of the heap as well. Instead, the algorithm maintains a buffer to store the incoming elements from the stream and merges the buffered elements into

L whenever the buffer is full. Specifically, the algorithm proceeds as follows.

1. Buffer the arriving elements into an array A . When A is full, sort the elements in A .
2. Merge A into L . More precisely, we scan A and L in parallel. In doing so, for each $v \in A$, we can find the smallest $v_i \in L$ such that $v_i > v$ and thus can compute the (v, g, Δ) tuple for v . During the merge, we also check each tuple in A or L to see if it is removable, and if so remove it (i.e., do not output it to the new L).
3. After the merging, flush the buffer and go back to step 1).

Since the size of the buffer is $\Theta(|L|)$, the cost of step (1) is $\Theta(|L| \log |L|)$, which is $O(\log |L|)$ per element amortized. The cost of step (2) is just $O(1)$ per element amortized. Thus, this variant has the same asymptotic update time (though amortized) as **GKAdaptive**. However, since sorting and merging are both much more cache-efficient than searching in a binary search tree and heap operations, this variant could be much faster in practice than **GKAdaptive**. We refer to this variant as **GKArray**.

2.2 q-digest

The q-digest [26] designed by Shrivastava et. al. was initially introduced as an algorithm for computing quantiles in a (distributed) sensor environment. It also applies to the streaming model. The algorithm is deterministic and assumes a fixed universe $[u]$.

The q-digest makes use of the dyadic structure of the universe $[u]$ described previously. This structure naturally corresponds to a complete binary tree with u leaves. Each leaf corresponds to an integer in $[u]$, while each node corresponds to a dyadic interval. We do not distinguish a node and its corresponding dyadic interval. Each node v is associated with a counter c_v , representing c_v elements from the stream in the dyadic interval of v . Initially, all counters are 0.

We use $lc(v)$ and $rc(v)$ to denote, respectively, the left child and right child of v . The following two invariants are maintained for any *internal* node v :

- (1) $c_v \leq \varepsilon n / \log u$;
- (2) $c_v + c_{lc(v)} + c_{rc(v)} > \frac{\varepsilon n}{\log u}$ if $c_{lc(v)} + c_{rc(v)} > 0$.

The first condition above ensures the accuracy of quantiles, and the second ensures that there are at most $O(\frac{1}{\varepsilon} \log u)$ nodes with nonzero counters. We say a node v is empty if $c_v = 0$, and it is full if $c_v = \varepsilon n / \log u$. Note that a full node may become non-full as n increases. As most tree nodes are empty, we only store the non-empty nodes and denote by Q the set of non-empty nodes.

To extract the ϕ -quantile, we sort all nodes (dyadic intervals) in Q based on the left endpoints of the intervals, breaking ties by putting smaller intervals first. Then, we find the first node v such that the sum of counters of v and all nodes before v are greater than ϕn . Finally, we return the right endpoint of v . It can be verified that this is an ε -approximate ϕ -quantile due to condition (1) above.

To merge two q -digests, we first add up the corresponding counters and then carry out a COMPRESS procedure [26] to make sure that condition (2) above is maintained. To adapt it to the streaming model, for each element in the stream, we simply increment the counter of its corresponding leaf, without enforcing condition (2). Then, we carry out the COMPRESS procedure whenever $|Q|$ doubles. Since COMPRESS takes $O(|Q| \log u)$ time [26], the amortized update cost per element is $O(\log u)$.

We see in our experiments that the $O(\log u)$ update time translates to a high cost in practice. This can be explained by observing that each element begins as a leaf in the structure, but the COMPRESS operation moves it up within the tree structure one step at a time until it comes to rest. Based on this observation we designed a variant called **FastQDigest**, following the discussion in [10]. The idea is to insert each new update directly in the tree structure where it would reside following a COMPRESS, without the lengthy search. We start with an empty Q . To insert an element x , we find the lowest ancestor of x in Q , say v (if Q is empty, we choose the root as v). If increasing c_v by 1 does not violate condition (1), we do so. Otherwise, we find the child of v that is also the ancestor of x (or x itself), set its counter to 1, and add it to Q . Finally, we call COMPRESS whenever n doubles. Since the accuracy of the algorithm only depends on condition (1), the correctness is still ensured.

Now, we observe that the set of non-empty nodes Q in this variant always form a connected subtree rooted at the root of the dyadic tree (which is not the case in the original version). This means that to find the lowest ancestor of x in Q , we can do a binary search on the path from x to the root. This can be done in $O(\log \log u)$ time, by storing Q in a hash table. But in the experiments, we observe substantial overhead with the hash table, so we used a simpler version where we replaced the hash table with a binary search tree (using `std::map`). We observe that the lowest ancestor of x in Q is exactly the innermost dyadic interval containing x . So we can store all nodes (dyadic intervals) of Q in the binary search tree, ordered by their left endpoints, breaking ties by putting longer intervals first. Now the innermost interval containing x is found by simply locating the predecessor of x , which is efficiently supported by this data structure.

COMPRESS is also more efficient in this variant, in time proportional to $|Q|$. Recall that all nodes in Q form a connected subtree. To compress Q , we try to move as many elements as possible from the counter of each node to its

ancestors, without violating condition (1). This can be done by a post-order traversal of Q .

2.2.1 Space and time analysis

After each COMPRESS, it is clear that all nodes with at least one non-empty child must be full; therefore, $|Q|$ is bounded by $O(\frac{1}{\varepsilon} \log u)$. Next, we show that between two consecutive COMPRESS operations, its size remains $O(\frac{1}{\varepsilon} \log u)$. Suppose n goes from n_0 to $2n_0$ between two consecutive COMPRESS operations. Since we only add a node to Q when its parent's counter is $\varepsilon n / \log u \geq \varepsilon n_0 / \log u$, there are at most twice as many of these nodes as those with counters greater than $\varepsilon n_0 / \log u$. There are at most $\frac{2n_0}{\varepsilon n_0 / \log u} = O(\frac{1}{\varepsilon} \log u)$ nodes of the latter type before we do the next COMPRESS, so the size of Q remains bounded by $O(\frac{1}{\varepsilon} \log u)$.

It takes time $O(\log |Q|) = O(\log \frac{1}{\varepsilon} + \log \log u)$ to insert an element into the data structure. The query time is also the same. Then, recall that we call COMPRESS whenever Q doubles, so it is invoked at most $O(\log n)$ times over the entire stream. Its amortized cost of $O(\frac{\log n \log u}{\varepsilon n})$ is negligible for n sufficiently large.

2.3 The randomized algorithm: Random

We now describe a randomized quantile algorithm, which can be seen as a simplified version of the one by Manku et al. [22]. It is also inspired by the algorithm by Agarwal et al. [1] that provides the mergeable property. We denote this algorithm as **Random**. It will report all quantiles within the specified error with constant probability.

Setting $h = \log \frac{1}{\varepsilon}$, $b = h + 1$ and $s = \frac{1}{\varepsilon} \sqrt{\log \frac{1}{\varepsilon}}$, **Random** maintains b buffers of size s each. Each buffer X is associated with a level $l(X)$.

Two buffers at the same level l can be merged into one buffer at level $l + 1$. To do so, in the sorted sequence of elements from both buffers, we randomly choose half of them: either those at odd positions, or those at even positions, each with probability $1/2$. The merged 2 buffers are then marked as empty.

Initially, all buffers are marked as empty. We set the active level $l = \max \left\{ 0, \left\lceil \log \frac{n}{s2^{h-1}} \right\rceil \right\}$. If there is an empty buffer X , we read the next $2^l s$ elements from the stream. For every 2^l elements, we randomly pick one and add it to X . Thus, X contains s sampled elements, becoming full, unless the stream is terminated. X is associated with level l . Whenever all buffers become full, we find the lowest level that contains at least 2 buffers, and merge 2 of them together.

In the end, the rank of an element v is estimated as $\hat{r}(v) = \sum_X 2^{l(X)} |\{i < v | i \in X\}|$, where X ranges over all non-empty buffers. A ϕ -quantile is reported as the element whose

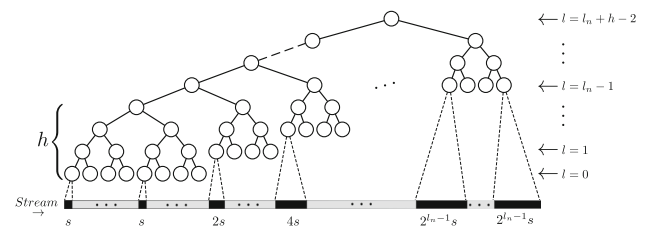


Fig. 1 Illustration of Random

estimated rank is closest to ϕn , which can be found using a binary search.

Figure 1 illustrates the algorithm. New elements of the stream arrive at the right of the figure. The algorithm can be understood in terms of a binary tree imposed over the stream. Each node in the tree corresponds to a buffer, and internal nodes are formed from the merger of their two children. Initially, leaf buffers are filled from s elements directly from the stream, but as the stream goes on, sampling is applied to fill the leaf buffers. There are 2^{h-1} leaf buffers at level 0, each storing s elements from the stream; for $1 \leq l < l_n$, there are 2^{h-2} leaf buffers at level l , each storing s elements sampled from $2^l s$ elements in the stream. There are 2^{h-2} non-leaf buffers at level l for any $1 \leq l \leq l_n$, and $2^{l_n+h-l-2}$ non-leaf buffers for $l_n + 1 \leq l \leq l_n + h - 2$.

2.3.1 Space and time analysis

Two buffers can be merged in $O(s)$ time, and the total number of merges is $O(n/s)$ throughout the entire data stream, which is amortized $O(1)$ for each update. Each buffer is sorted when it just becomes full, which can be done in $O(s \log s)$ time, so $O(\log s)$ per update amortized. Hence, the amortized update time is $O(\log s) = O(\log \frac{1}{\varepsilon})$.

The space bound is simply $bs = O(\frac{1}{\varepsilon} \log^{1.5} \frac{1}{\varepsilon})$.

2.3.2 Error analysis

We show that with constant probability, this algorithm finds all quantiles correctly.

Since our analysis will focus on the asymptotics, we assume that n/s is a power of 2, which means that when the stream terminates, l has been just increased by 1 and becomes $l_n = \log(\frac{n}{s}) + 2$. In order to simplify the proof, at this point we merge all the buffers into one, whose level is $l_n + h - 2 = \log(n/s)$. Note that this operation can only increase the error.

We first consider the all-quantile guarantee. If the estimated ranks of all the $1/\varepsilon - 1$ elements that rank at $\varepsilon n, 2\varepsilon n, \dots, (1 - \varepsilon)n$ are correct (i.e., with at most additive εn error), then all the quantiles can be answered correctly. By the union bound, it suffices to ensure that each rank is correct with probability (strictly) greater than $1 - \varepsilon$.

When the algorithm estimates the rank of any element, the error comes from two sources: random sampling and random merging. Clearly, the expected error of each type is zero, so the estimator is unbiased. Now we analyze the probability that the error is larger than εn . For the random sampling part, consider any sampled element at level l , which has been chosen from 2^l elements, so the error is between -2^l and 2^l . By Hoeffding's inequality, the probability of the absolute value of their sum exceeding εn is at most

$$\exp\left(-\frac{2(\varepsilon n)^2}{\sum_{\text{leaf buffer } X} 4^{l(X)} s}\right) = \exp\left(-\Theta\left(\varepsilon^2 2^h s\right)\right), \quad (1)$$

since the summation over X is dominated by the contribution from the highest level, where $l(X) = \log n/s$. Note that this probability can be made to be smaller than ε^c for any constant c by increasing h by a constant factor.

Next consider the error from the random merging. Merging 2 buffers at level l may contribute an error between -2^l and 2^l . Again by Hoeffding's inequality, the probability that the total error exceeds εn is bounded in terms of the sum of the squares of the absolute errors (also dominated by the contribution of the highest level), as

$$\exp\left(-\frac{2(\varepsilon n)^2}{\sum_{\text{non-leaf buffer } X} 4^{l(X)} s}\right) = \exp\left(-\Theta\left(\varepsilon^2 s^2\right)\right). \quad (2)$$

With $s = \Theta\left(\frac{1}{\varepsilon} \sqrt{\log\left(\frac{1}{\varepsilon}\right)}\right)$, this can be made to ε^c for any constant c .

Finally, we consider how to achieve the one-quantile guarantee. As observed earlier, the failure probability (1) from the random sampling can also be made to be ε^c for any constant c without affecting the asymptotic result, so it suffices to only consider the failure probability from the random merging in (2). To achieve the one-quantile guarantee, this just needs to be a constant, which requires $s = O(1/\varepsilon)$, leading to the space bound of $O(\frac{1}{\varepsilon} \log \frac{1}{\varepsilon})$ as claimed in Table 1.

2.4 The MRL99 algorithm

As mentioned, the algorithm **Random** can be seen as a simplified version of the one by Manku et al. [22], which we denote as **MRL99**. Compared with **Random**, **MRL99** has the following complications. (1) The parameters b , h and s are determined by solving a complicated optimization problem, whereas in our case, they are set easily. (2) In addition to level, each buffer X is also associated with a weight. In our case, the weight is always $2^{l(X)}$ so it is implicit, but this may not be the case in **MRL99**. (3) A more complex merging

procedure is used that may merge buffers of different weights together and may merge more than two buffers at a time. In **Random**, we only merge 2 at a time and they must have the same weight.

Due to these complex procedures and the delicacy in setting the parameters, **MRL99** is very difficult to analyze. As a result, Manku et al. [22] only gave a pessimistic $O\left(\frac{1}{\varepsilon} \log^2 \frac{1}{\varepsilon}\right)$ bound, which is not interesting by today's standard since it can also be obtained by simply running $O(\log \frac{1}{\varepsilon})$ instances of the GK algorithm on multiple random samples of the stream. Nevertheless, the practical behavior of the algorithm should be very competitive.

2.5 Random over sliding windows

In this section, we consider a variant of the cash register streaming model, in which only the last W elements are considered. This is usually known as the *sliding window* streaming model.

The only known quantile algorithm in the sliding window model is due to Arasu and Manku [3], which maintains a dyadic structure over the sliding window of length W as shown in Fig. 2. The dyadic structure consists of $L = \log \frac{4}{\varepsilon}$ levels. On the l th level, the elements are partitioned into blocks of size $b_l = \frac{\varepsilon W}{4} 2^l$, and they run a standard quantile algorithm for each block with error parameter $\varepsilon_l = \frac{\varepsilon}{L} 2^{L-l}$. Note that with these parameters, the absolute error from each block is the same, which is $\varepsilon_l b_l = O(\varepsilon W/L)$. Because at any time, the sliding window can always be decomposed into at most L blocks (ignoring possibly one boundary block on level 0, which has only $\varepsilon W/4$ elements), the total error is $O(\varepsilon W)$, as desired.

The blocks can be classified into three types: *expired*, *active*, and *under construction*, as illustrated in Fig. 2. No data structures are needed for the expired blocks. The active blocks will not receive any new elements, so when the quantile algorithm for each block finishes, we can extract the $(i \cdot \varepsilon_l)$ -quantiles for $i = 1, 2, \dots, 1/\varepsilon$ and discard the auxiliary data structures. Thus, each active block only needs $O(1/\varepsilon_l)$ space. The blocks under construction still need

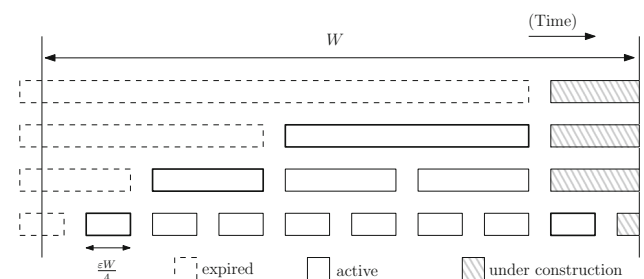


Fig. 2 Dyadic structure over sliding window W consisting of four levels

quantile algorithms running with their auxiliary data structures. By plugging in the GK algorithm for each block, the total space is $O\left(\frac{1}{\varepsilon} \log \frac{1}{\varepsilon} \log(\varepsilon n)\right)$; by taking a random sample of size $O\left(\frac{1}{\varepsilon^2} \log \frac{1}{\varepsilon}\right)$ and then running the GK algorithm on the sample, the space is $O\left(\frac{1}{\varepsilon} \log^2 \frac{1}{\varepsilon}\right)$, which holds for both the all-quantile and the one-quantile guarantee.

Obviously, we can also use our **Random** algorithm for each block. An interesting observation is the following. GK is a deterministic algorithm that gives an error of $\varepsilon_l b_l = \varepsilon W/L$ for each block, so the total error is simply L times that, i.e., $L \cdot \varepsilon_l b_l = \varepsilon W$. However, the **Random** algorithm gives an unbiased result with error $\pm \varepsilon_l b_l$, which, when added together across L blocks, can potentially cancel out. It turns out we can afford to use a larger $\varepsilon_l = \frac{\varepsilon}{\sqrt{L}} 2^{L-l}$, which means that the space needed by each block can be smaller by a $\sqrt{L} = O\left(\log^{0.5} \frac{1}{\varepsilon}\right)$ factor.

Here is a more formal analysis. Recall that a quantile query on the sliding window can be decomposed into L quantile queries. The **Random** algorithms answer each query with an unbiased result with error $\pm \varepsilon_l b_l = O(\varepsilon W/\sqrt{L})$. By Hoeffding's inequality, the probability that the total error exceeds εW is

$$\exp\left(-\frac{2(\varepsilon W)^2}{O(L(\varepsilon W/\sqrt{L})^2)}\right) \leq \frac{1}{2}. \quad (3)$$

Note that the constant $\frac{1}{2}$ can be made arbitrarily small. The total space occupied by all the active blocks is thus

$$\sum_{l=0}^L \frac{1}{\varepsilon_l} \cdot 2^{L-l} = \frac{1}{\varepsilon} L^{1.5} = O\left(\frac{1}{\varepsilon} \log^{1.5} \frac{1}{\varepsilon}\right).$$

The total space occupied by the blocks under construction (for the one-quantile guarantee) is

$$\sum_{l=0}^L \frac{1}{\varepsilon_l} \log \frac{1}{\varepsilon_l} = O\left(\frac{1}{\varepsilon} \log^{1.5} \frac{1}{\varepsilon}\right).$$

This improves upon the algorithm of Arasu and Manku [3] for the one-quantile guarantee.

Unfortunately, for the all-quantile guarantee, we need the probability (3) to be $O(\varepsilon)$, which does not allow us to use a larger ε_l . However, this is mostly due to the looseness in our analysis; in Sect. 4, we will perform an experimental study comparing the two versions of sliding window quantile algorithms.

3 Turnstile algorithms

Recall that all algorithms in the turnstile model build upon the dyadic structure over the universe $[u]$ as described in Sect. 1,

and use a frequency estimation sketch for each level. Known turnstile quantile algorithms only differ in the sketches they choose to use. Over a stream of updates with both insertions and deletions of elements, a frequency estimation sketch should be able to return an estimate of the frequency of any given element x . Note that when used in level i in the dyadic structure (the bottom level is level 0), an “element” is actually a dyadic interval of length 2^i , and the frequency estimation sketch summarizes a reduced universe $[u/2^i]$. Thus, for an integer x in the stream, we take its first $\log(u) - i$ bits to map it to level i . Finally, it is obvious that if the reduced universe size $u/2^i$ is smaller than the sketch size, we should maintain the frequencies exactly, rather than using a sketch.

In the turnstile model, we use n to denote the number of elements currently remaining, which is at most the stream length.

3.1 Random subset sum

Gilbert et al. [13] were the first to consider the quantile problem in the turnstile model and designed the *random subset sum* sketch as a frequency estimation sketch to be used in the dyadic structure. However, it results in an overall size of $O\left(\frac{1}{\varepsilon^2} \log^2 u \log\left(\frac{\log u}{\varepsilon}\right)\right)$: the dependence on ε is much higher than subsequent methods. Moreover, its update time is proportional to its size. Our implementation experience with this algorithm confirmed these properties, with its update being 10^3 – 10^6 times slower than the other algorithms. So we exclude it from further experimental evaluation.

3.2 Count-Min sketch

The Count-Min sketch consists of an array C of $w \times d$ counters, all initialized to 0. For each of the d rows of the array, it uses a pairwise independent hash function $h_i : [u] \rightarrow [w]$ that maps the elements in the universe to the w counters in this row. To insert/delete an element x in the sketch, we add/subtract 1 from the counter $C[i, h_i(x)]$, for $i = 1, \dots, d$. To estimate the frequency of x , we return $\min_i C[i, h_i(x)]$.

It has been shown [7] that if $w = O(1/\varepsilon)$ and $d = O(\log \frac{1}{\delta})$, then the estimate has at most εn error with probability at least $1 - \delta$. To use this in the dyadic structure, we only allow $\varepsilon n / \log u$ error from each level, so we use an error parameter $\varepsilon' = \varepsilon / \log u$ in each sketch. To find a quantile, we do a binary search with $\log u$ probes, where each probe involves $\log u$ queries to the sketches. We also want all $1/\varepsilon$ quantiles to be correct with constant probability, so a union bound implies that we need to set the failure probability of each sketch to $\delta' = \Theta(\varepsilon / \log^2 u)$. This leads to an overall size of $O\left(\frac{1}{\varepsilon} \log^2 u \log\left(\frac{\log u}{\varepsilon}\right)\right)$. To process an update in the stream, we need to update $\log u$ sketches, one from each level, while updating each sketch requires updating all its d rows. So the total update time is $O(\log u \log(\frac{\log u}{\varepsilon}))$. These

results are stated in [7]. We denote this algorithm as DCM (Dyadic Count-Min).

3.3 DCS: Dyadic Count-Sketch

We propose to use the Count-Sketch [5] for frequency estimation in the dyadic structure. The Count-Sketch consists of an array C of $w \times d$ counters. For each row i , it uses a pairwise independent hash function $h_i : [u] \rightarrow [w]$ that maps the elements in the (reduced) universe to the w counters in this row, as well as a 4-wise independent hash function $g_i : [u] \rightarrow \{-1, +1\}$ that maps each element to -1 or $+1$ with equal probability. To insert/delete an element x in the sketch, for each row i , we add/subtract $g_i(x)$ to $C[i, h_i(x)]$. To estimate the frequency of x , we return the median of $g_i(x) \cdot C[i, h_i(x)]$, $i = 1, \dots, d$ (assuming d is odd).

By setting $w = O(1/\varepsilon)$ and $d = O(\log \frac{1}{\delta})$, the Count-Sketch returns an estimate with more than εn error with probability at most δ , which is the same as the Count-Min sketch. However, we observe another property of the Count-Sketch that makes it appealing for the quantile problem, that it produces an unbiased estimator. Since we add up the estimates from $\log u$ sketches in the dyadic structure, it is likely that some of the positive and negative errors will cancel each other out, leading to a more accurate final result. Below we give a new analysis showing that this intuition in fact leads to an asymptotic improvement over using the Count-Min sketch for the quantile problem (DCM).

3.3.1 Analysis

Below we prove that DCS can return all ε -approximate quantiles with constant probability using space $O(\frac{1}{\varepsilon} \log^{1.5} u \log^{1.5}(\frac{\log u}{\varepsilon}))$.

Consider the estimators $Y_i = g_i(x) \cdot C[i, h_i(x)]$, $i = 1, \dots, d$. Each Y_i is clearly unbiased, since $g_i(x)$ maps to -1 or $+1$ with equal probability. Let Y be the median of the Y_i 's. The median of independent unbiased estimators is not necessarily unbiased, but if each estimator also has a symmetric pdf, then this is the case. This result seems to be folklore. In our case, each Y_i has a symmetric pdf, so Y is still unbiased.

Using the same argument as for the Count-Min sketch, we have

$$\Pr[|Y_i - E[Y_i]| > \varepsilon n] < 1/4.$$

Since Y is the median of the Y_i 's, by a Chernoff bound, we have

$$\Pr[|Y - E[Y]| > \varepsilon n] < \exp(-O(d)).$$

Now consider adding up $\log u$ such estimators; the sum must still be unbiased. By the union bound, the probability that every estimate has at most εn error is at least $1 - \exp(-O(d)) \cdot \log u$. Conditioned upon this event happening, we can use Hoeffding's inequality to bound the probability that the sum of $\log u$ such (independent) estimators deviate from its mean by more than t as

$$2 \exp\left(-\frac{2t^2}{(2\varepsilon n)^2 \log u}\right).$$

We see that if we set $t = \Theta(\varepsilon n \sqrt{\log u})$, this probability will be a constant. This means that, summing over the $\log u$ levels, the error only grows proportionally to $\sqrt{\log u}$.

To make this bound rigorous, we must ensure that all quantiles are correct with constant probability. So each such sum should fail with probability no more than $\varepsilon / \log u$. Thus, we set $t = \Theta\left(\varepsilon n \sqrt{\log u \log(\frac{\log u}{\varepsilon})}\right)$. In addition, we need

to choose $d = \Theta(\log(\frac{\log u}{\varepsilon}))$ to ensure that the prerequisite condition holds with probability at least $1 - \varepsilon / \log u$. Finally, to get εn error in the end, we use a parameter $\varepsilon' = \varepsilon / \sqrt{\log u \log(\frac{\log u}{\varepsilon})}$ in the sketches (i.e., $w = 1/\varepsilon'$). Thus, the total space of DCS is $w \cdot d \cdot \log u = O\left(\frac{1}{\varepsilon} \log^{1.5} u \log^{1.5}(\frac{\log u}{\varepsilon})\right)$, and its update time and query time is $d \cdot \log u = O(\log u \log(\frac{\log u}{\varepsilon}))$, as claimed in Table 1.

3.4 Post-processing

All the quantile algorithms make use of the dyadic structure and use an independent frequency estimation sketch for each level. However, the true frequencies across different levels are *not* independent. Consider the toy example in Fig. 3, which shows a dyadic structure on a tiny universe $\{0, 1, 2, 3\}$. For each node v in this binary tree, DCS returns an unbiased estimator Y_v for the number of elements in the corresponding interval. Writing x_v to denote the true frequency at node v , we have the prior knowledge that $x_1 = x_2 + x_3$, $x_2 = x_4 + x_5$ and $x_3 = x_6 + x_7$. However, it is very unlikely that $Y_1 = Y_2 + Y_3$ or $Y_2 = Y_4 + Y_5$. The question is thus, can we use this prior knowledge to improve the accuracy of the Y_i 's (which in turn leads to better accuracy for quantile approximation). The

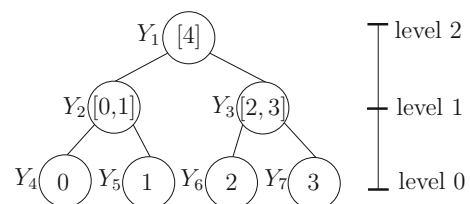


Fig. 3 Binary tree corresponding to the dyadic structure on the universe $\{0, 1, 2, 3\}$

answer is yes, at least on this toy example. Let us assume that all the Y_i 's are independent (actually, pairwise independence suffices), and have the same variance σ^2 . Consider Y_2 . If we set $Y'_2 = Y_1/2 + (Y_2 - Y_3)/3 + (Y_4 + Y_5 - Y_6 - Y_7)/6$, it can be checked that $E[Y'_2] = E[Y_2]$ and $\text{Var}(Y'_2) = \frac{7}{12}\sigma^2$, namely Y'_2 is still an unbiased estimator of x_2 but with a smaller variance than the original estimator. The reason we can achieve this improvement is that the other estimators include information about x_2 , which can be used in conjunction with Y_2 . For instance, $Y_4 + Y_5$ and $Y_1 - Y_6 - Y_7$ are unbiased estimators of x_2 (from independent estimators), and these can be combined to reduce the variance.

More questions naturally follow: Is this the best estimator of x_2 ? Is there a principled approach? Can we compute the improved estimators efficiently? These are the questions we address in this section.

3.4.1 Ordinary least squares

We can formalize the problem as follows. Let $x = (x_i)$ be a vector of hidden values. We are given a vector of observations $y = (y_i)$, where each y_i is an unbiased estimator of some linear combination of the x_i 's. This can be succinctly expressed in a matrix form:

$$y = Ax + \delta,$$

where $\delta = (\delta_i)$ is a vector of pairwise independent random variables with mean 0, and $\text{Var}(\delta_i) = \sigma_i^2$. In our case, the x_i 's are the true frequencies at the leaves of the binary tree T corresponding to the dyadic structure, the y_i 's are the estimators returned by the Count-Sketch at every node of T , and A is a 0–1 matrix that encodes which x_i 's are covered by each y_i .

The problem is thus to obtain the best estimates for the x_i 's. Once the x_i 's are known, the values at internal nodes of T can be computed easily. This is exactly the *ordinary least squares* (OLS) problem, and the *best linear unbiased estimator* for x is the vector x^* such that $\sum_i (y_i - A_i x^*)^2 / \sigma_i^2$ is minimized. According to the Gauss–Markov theorem [25], the variance of any linear combination of the x_i^* 's is also minimized, so it exactly serves our purpose, since the rank of any element is the linear combination of some x_i 's.

However, one technicality in our case is that some of the σ_i 's are 0: for some nodes high in the tree, we record the exact frequencies when this is more space efficient than using a sketch. This means that for some k , we have $\sigma_i = 0$ for $i = 1, \dots, k$. Thus, the modified problem formulation is as follows.

Definition 1 Given a vector $x = (x_i)$ of unknowns, a vector $y = (y_i)$ of observations, an $m \times n$ matrix A , and an integer k , such that

$$\begin{cases} y_i = A_i x, & i \leq k \\ y_i = A_i x + \delta_i, & i > k, \end{cases} \quad (4)$$

where A_i is the i th row of A . The δ_i 's are pairwise independent random variables with mean 0 and variance $\sigma_i^2 > 0$. A vector $x^* = (x_i^*)$ is the *best linear unbiased estimator* (BLUE) for x if $y_i = A_i x^*$ for all $i \leq k$ and $\sum_{i>k} (y_i - A_i x^*)^2 / \sigma_i^2$ is minimized.

The general method for solving this problem is either to eliminate the y_i 's for $i \leq k$ by Gauss elimination, or use the method of Lagrange multipliers [19]. However, these both would take $O(u^3)$ time as A in our case is a $(2u - 1) \times u$ matrix, where u is the universe size! In the rest of this section, we will exploit the special properties of our setting and develop much more efficient algorithms.

3.4.2 Truncating the tree

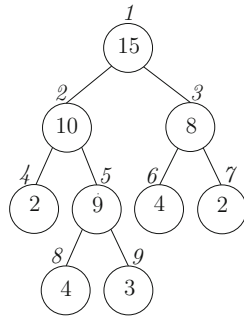
Our first observation is that we should not work on the entire dyadic structure. Because we can tolerate an error of εn , any interval that has less than εn weight can be safely discarded. More precisely, we extract a truncated binary tree $\hat{T}$ from the DCS as follows. Starting from root we traverse the dyadic structure top-down. For each node (interval), we estimate its frequency from the Count-Sketch. If it is larger than εn , we recursively visit its children; otherwise, we skip this node as well as its subtree. In “Size of the truncated tree $\hat{T}$ ” section of Appendix, we show that the size of the truncated tree $\hat{T}$ is only $O(\frac{1}{\varepsilon} \log u)$ in expectation. In our implementation, in order to have better accuracy, we set the truncating threshold to $\eta \varepsilon n$ for some small constant η and experimentally tune the parameter η to achieve a desired trade-off between accuracy and cost.

Even after truncating the tree, the cubic running time using the standard OLS method is still too expensive. By exploiting the special properties of the tree structure, Li et al. [20] designed an algorithm to compute the BLUE in linear time. However, their algorithm can only work on a perfectly balanced tree. In our case, $\hat{T}$ can be very unbalanced, which is especially the case on skewed distributions. Furthermore, their algorithm cannot handle the case where some σ_i 's are 0. Below we present our algorithm that resolves these issues.

3.4.3 The algorithm

First, we decompose the tree $\hat{T}$ into subtrees such that the only node with an exact frequency is the root. It is clear that an exact node “shields” the influence of its subtree from other parts of tree, so each subtree can be handled separately.

Let T_r be such a tree with root r . We use $w \prec v$ to denote that w is a leaf below node v . For a node v , let $\text{lpath}(v)$ be the set of nodes on the path from v to the leftmost leaf

Fig. 4 Binary tree

below v , $\text{anc}(v)$ the set of all ancestors of v (including v), and $\text{parent}(v)$ the parent node of v . For each node $v \in T_r$, we are given y_v , which is an unbiased estimator of $x_v = \sum_{w < v} x_w$, with variance σ_v^2 , and we wish to compute the BLUE x_v^* for each v .

An important technique to handle an unbalanced tree is to introduce a *weight* λ_v for each node v . Intuitively, the weight of a node measures how important it is for computing the BLUE for the whole system. When the tree is perfectly balanced, the weights are the same for all the nodes on the same level. But on an unbalanced tree, the weights will depend on both σ_v and the structure of the tree. Consider the example in Fig. 4, where the number inside each node is the estimated frequency y_v . Here we assume $\sigma_i^2 = 2$ for all i except that $\sigma_1^2 = 0$. Nodes 4 and 5 are on the same level and have the same variance. However, node 4 has no children, while node 5 has two children whose sum is another estimate of the true frequency at node 5. Thus, the importance of the estimate at node 5 itself should be discounted relative to that at node 4. This intuition is captured quantitatively by the following equations:

$$\begin{cases} \lambda_v = \sum_{w < v} \lambda_w, & \text{for all internal nodes } v; \\ \pi_{\text{left child of } v} = \pi_{\text{right child of } v}, & \text{for all internal nodes } v, \end{cases} \quad (5)$$

where $\pi_v = \sum_{w \in \text{path}(v)} \lambda_w / \sigma_w^2$.

Note that if T_r has τ leaves, hence $\tau - 1$ internal nodes, then there are $2\tau - 1$ weights and (5) has $2\tau - 2$ equations in total. Thus, (5) does not uniquely determine the weights, but they only differ by scaling. This will not be a problem since the weights will only measure the relative importance of the nodes. For convenience, we add the constraint $\lambda_r = 1$ to make the weights uniquely defined. This way, the weights can be computed by solving the system of linear Eq. (5), and those for the example in Fig. 4 are given in Table 2.

We can use a bottom-up traversal of T_r to efficiently solve (5) in linear time. We start from any node that is just above the leaf level. Let v be such a node with two children leaves w_1 and w_2 . We have two linear equations at v involving

Table 2 Computing the BLUE

Node	λ	π	Z	F	x^*
1	1	–	419/62	0	15
2	15/31	12/31	243/62	4.47	8.94
3	16/31	12/31	88/31	3.03	6.06
4	9/31	9/62	54/31	–	1.16
5	6/31	9/62	135/62	8.36	7.77
6	8/31	4/31	48/31	–	4.04
7	8/31	4/31	40/31	–	2.03
8	3/31	3/62	69/62	–	4.38
9	3/31	3/62	33/31	–	3.38

$$\Delta = 59/62$$

3 unknowns $\lambda_v, \lambda_{w_1}, \lambda_{w_2}$. We solve them and obtain two relationships $\lambda_{w_1} = \alpha_{w_1} \lambda_v$, and $\lambda_{w_2} = \alpha_{w_2} \lambda_v$ for some $\alpha_{w_1}, \alpha_{w_2}$. We also have $\pi_v = \lambda_{w_1} / \sigma_{w_1}^2 + \lambda_{w_2} / \sigma_{w_2}^2 = \beta_v \lambda_v$ for some β_v . After this, we mark v as “done” and move on to any other node whose children are both done. In general, when we reach a node v with children u_1 and u_2 , we will have inductively obtained $\pi_{u_1} = \beta_{u_1} \lambda_{u_1}, \pi_{u_2} = \beta_{u_2} \lambda_{u_2}$ for some β_{u_1}, β_{u_2} . We can then solve the 2 equations at v involving 3 unknowns $\lambda_v, \lambda_{u_1}, \lambda_{u_2}$ and obtain relationships $\lambda_{u_1} = \alpha_{u_1} \lambda_v, \lambda_{u_2} = \alpha_{u_2} \lambda_v$, as well as $\pi_v = \pi_{u_1} + \lambda_v / \sigma_v^2 = \beta_v \lambda_v$, for some $\alpha_{u_1}, \alpha_{u_2}, \beta_v$. After this traversal, we will have discovered the relationship between any λ_v and $\lambda_{\text{parent}(v)}$, and plugging in $\lambda_r = 1$ will yield all the λ_v ’s and π_v ’s in linear time.

In order to compute the x_i^* ’s efficiently, we need some more auxiliary variables. For each leaf $w \in T_r$, let $Z_w = \lambda_w \sum_{z \in \text{anc}(w) \setminus \{r\}} y_z / \sigma_z^2$, and for any internal node v , $Z_v = \sum_{w < v} \lambda_w Z_w$. For any node v except the root r , let $F_v = \sum_{w \in \text{anc}(v) \setminus \{r\}} x_w^* / \sigma_w^2$. Finally, set $\Delta = (Z_r - y_r \pi_r \text{’s left child}) / \lambda_r$. The values of these auxiliary variables for the example in Fig. 4 are given in Table 2.

Using these auxiliary variables, we are able to obtain the following equations with respect to the x_i^* ’s (proof given in “Constraints on the x_i^* ’s” section of Appendix):

$$\begin{cases} x_r^* = y_r, \\ x_v^* = (Z_v - \lambda_v F_{\text{parent}(v)} - \lambda_v \Delta) / \pi_v, & \text{for all nodes } v \neq r. \end{cases} \quad (6)$$

Note that (6) has exactly $2\tau - 1$ unknowns and $2\tau - 1$ equations. We can then efficiently solve (5) and (6) in three traversals of T_r , as follows:

1. *Top-down traversal to compute a temporary Z'* : Initialize $Z'_r = 0$; for any other v , set $Z'_v = Z'_{\text{parent}(v)} + y_v / \sigma_v^2$ recursively. Note that after this traversal, for any leaf w , we have computed $Z_w = \lambda_w Z'_w$.

2. *Bottom-up traversal to compute Z* : With the values of Z_w 's at the leaves, we can easily compute all the Z_v 's a bottom-up traversal.
3. *Top-down Traversal to compute F, x^** : We first compute $\Delta = (Z_r - y_r \pi_s) / \lambda_r$, and initialize $F_r = 0$, $x_r^* = y_r$. During this top-down traversal, we compute $F_v = F_{\text{parent}(v)} + x_v^* / \sigma_v^2$ for every v . Meanwhile, each x_v^* can be computed from (5).

It is easy to see that the algorithm takes time linear in the size of the truncated tree $\hat{T}$, which is $O(\frac{1}{\varepsilon} \log u)$ in expectation.

3.4.4 Discussion

There are a few issues to discuss with respect to our post-processing algorithm. First, the OLS framework requires pairwise independence among the estimators y_v . This is not completely true in our setting. Two nodes on different levels are clearly independent, as they are returned from independent Count-Sketches. However, two nodes on the same level are not. Note that although the Count-Sketch uses a pairwise independent hash function to distribute elements into counters, the counters themselves are *not* pairwise independent. Nevertheless, we have analyzed the covariance of any two counters and shown that it is much smaller than the variance (details in ‘‘Covariance analysis’’ section of Appendix). Therefore, we argue that the OLS framework is still suitable to apply.

The second issue is that our algorithm needs the variance σ_v^2 for each v . Conveniently, the Count-Sketch itself actually provides a good estimator for this variance, which is simply the sum of all the counters squared in a row [2]. However, when we use multiple rows and return the median estimator, the variance does not follow so easily. Nevertheless, our algorithm is not affected if all the σ_v^2 's are reduced by the same factor, so we use the variance of one row of the sketch as a good empirical approximation.

3.4.5 An online version

The post-processing method, as the name indicates, is designed to improve the accuracy of the sketch in an offline setting. Because its running time is $O(\frac{1}{\varepsilon} \log u)$, we cannot afford to use it after each update. However, there is a simple trick that can turn it into an online version without having a major impact on the space and time costs of the original DCS algorithm.

Recall that the post-processing method first extracts a truncated binary tree $\hat{T}$ from the sketch, where each node u of $\hat{T}$ is associated with a counter for the total number of elements covered by u . All nodes with counters greater than $\eta \varepsilon n$ are

kept in the tree. Then, it uses OLS to improve the accuracy of these counters.

Our idea to turn this into an online algorithm is the following. After we have obtained an improved binary tree $\hat{T}$, we keep it together with the DCS sketch. Quantile queries are answered online with $\hat{T}$. Each update is applied to the DCS sketch as before, but at the same time, we also update the affected counters in $\hat{T}$. This way, $\hat{T}$ always reflects the current state of the data at any point in time.

It may appear that we can do this forever, which is too good to be true. Indeed, a deletion has no impact on the accuracy of $\hat{T}$. Insertions will not affect the accuracy of the counters, either, but they may increase some counters that are not kept in $\hat{T}$. Recall that we should keep all counters above $\eta \varepsilon n$ in $\hat{T}$, but when some other counter passes the threshold due to many insertions, we may not know. Therefore, we adopt the following strategy. Whenever a leaf node in $\hat{T}$ has its counter exceeding $2\eta \varepsilon n$, we discard the current $\hat{T}$ and build a new one from the DCS sketch using the post-processing method. We call this version of the algorithm **vPost**.

Now we analyze the space and time costs of **vPost**. The running time of each invocation of the post-processing method is $O(\frac{1}{\varepsilon} \log u)$. However, we invoke it once for every $\eta \varepsilon n$ or more updates, which is $O(\frac{1}{\varepsilon^2 n} \log u)$ amortized. This is $O(1)$ for n sufficiently large. In addition, this is assuming the worst-case scenario in which all updates are insertions and all insertions fall under one leaf node in $\hat{T}$. In practice, we expect much less invocations. The tree itself takes space $O(1/\varepsilon)$, which is much smaller than the DCS sketch itself, so **vPost** has very little impact on the space and time costs of the DCS sketch, while returning more accurate results thanks to the post-processing method.

4 Experiments

4.1 Setup

We implemented all algorithms in C++, compiled with GCC. The executables were tested under Linux 2.6.18 on a machine with a 3GHz CPU, 6MB CPU cache and 16GB memory.

4.1.1 Data sets

We used 2 real data sets and 12 synthetic data sets in the experiments. The first real data set is the MPCAT-OBS data set, which is an observation archive available from the Minor Planet Center.² We used the optical observation records from 1802 to 2012. The records are ordered by the timestamp,

² <http://www.minorplanetcenter.net/iau/ecs/mpcat-obs/mpcat-obs.html>.

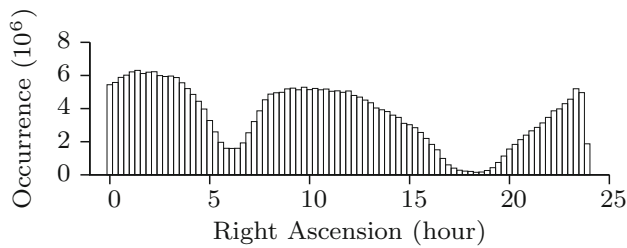


Fig. 5 Distribution of MPCAT-OBS

and we feed the right ascensions³ as a stream to the algorithms. The stream values appear to arrive randomly overall, but consist of chunks of ordered data of various lengths. This is because an observatory usually traces a planet continuously in a session and then moves on to other planets. The right ascension is not uniformly distributed, as shown in Fig. 5. This data set contains 87,688,123 records, and the right ascensions are integers ranging from 0 to 8,639,999. The second real data set is the terrain data for the Neuse River Basin,⁴ which contains LIDAR points measuring the elevation of the terrain. This data set contains about 100 million points.

In order to study how different data characteristics affect the algorithms' performance, we also generated 12 synthetic data sets with different sizes (10^7 – 10^{10}), universe sizes (2^{16} – 2^{32}), distributions (uniform and normal with different variances), and order (random and sorted). Further details are given in context. Note that we know that certain factors do not affect certain algorithms, due to their definition. For example, the universe size and distribution should not affect any comparison-based algorithms; the stream order should not affect (the space and accuracy of) the turnstile algorithms; and the stream length should not affect q-digest and the turnstile algorithms.

4.1.2 Measures

We measure the algorithms along the following dimensions: *Space* is one of the most important measures for streaming algorithms. We report space usage in bytes, where every element from the stream, counter, or pointer consumes 4 bytes. When an algorithm uses auxiliary data structures such as a binary tree or a hash table, the space needed by these internally is carefully accounted for. For algorithms whose space usage changes over time, we measured the maximum space usage.

³ Right ascension is an astronomical term used to locate a point (a minor planet in this case) in the equatorial coordinate system.

⁴ <http://www.ncfloodmaps.com>.

Update time is as important as space, if not more so, as it translates to the throughput of the streaming algorithm. Prior empirical studies have overlooked this issue [15, 21]; more recent works on other streaming problems have included time as a main consideration [6]. In our experiments, we measured the average wall clock processing time per element in the stream. In some cases, it is important to bound the worst-case time per element, and some algorithms periodically use a slower pruning procedure (e.g., a COMPRESS or merge step). We note that standard de-amortization techniques, such as use of buffering, can be adopted to avoid blocking operations.

Accuracy is the third factor we measure: We want to understand the accuracy–space and accuracy–time trade-offs. There are some technical subtleties in measuring the error. The error parameter ε used by the algorithms controls the accuracy, but it is not suited for use as the measure of empirical accuracy for two reasons. First, the error analysis usually considers worst-case input and may be loose: The actual error could be substantially better, and second, the deterministic algorithms provide an ε -error guarantee, while the randomized ones give such a guarantee only probabilistically, so it is not a fair comparison. Therefore, in our experiments, we measure the observed errors, and used the following two error metrics.

We first extract the ϕ -quantiles for $\phi = \varepsilon, 2\varepsilon, \dots, (1 - \varepsilon)$. For each ϕ -quantile extracted, we compute its true rank from the data, and take its difference from ϕn , divided by n . From all these errors, we take the maximum and average values. The former is exactly the *Kolmogorov–Smirnov divergence* between the true CDF and that of the extracted quantiles, while the latter is determined by the *total variation distance* of the two CDFs, both of which are standard statistical distances between distributions. There is some ambiguity over the rank of elements which appear multiple times in the data. We favor the algorithms, so that the rank of such items is taken as an interval. We compute the error as the difference from ϕn to the closer interval endpoint, or 0 if ϕn is contained within the interval.

Query time Finally, the time to answer a quantile query is also a meaningful measure. However, for almost all the algorithms, the query time turns out to be almost the same as the update time, which agrees with our analysis. So we will not show the query times explicitly. One exception is DCS after the post-processing. Since after the post-processing, we only need to query the binary tree $\hat{T}$ without having to examine the DCS sketch, so the query time is only $O(\log u)$, which is faster than the update time.

Thus, in total we make five measurements (space, time, ε , actual maximum error, actual average error) for each algorithm in each experiment. For randomized algorithms, we repeat the algorithm 100 times and take the average. For space reasons, we present a selection of most representative

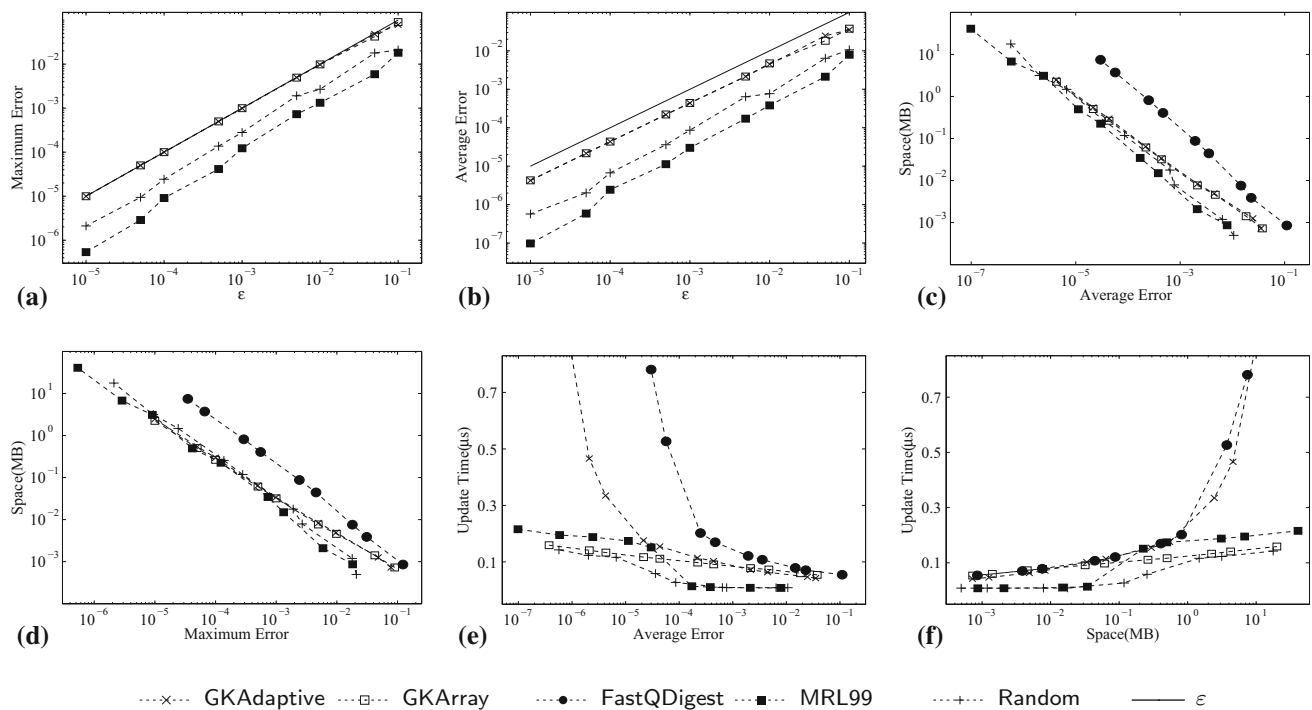


Fig. 6 Results on MPCAT-OBS. **a** Maximum error- ϵ , **b** average error- ϵ , **c** space-average error, **d** space-maximum error, **e** time-average error, **f** time-space

results in this paper; the full comparison across all 9 algorithms and 5 measurements over 14 real and synthetic data sets can be explored (anonymously) through an interactive interface at <http://quantiles.github.com>. Below, all results are on the MPCAT-OBS data set unless specified otherwise.

4.2 Results on cash register algorithms

4.2.1 ϵ versus actual error

Figure 6a, b shows how the actual errors of the algorithms deviate from the given ϵ parameter. All the deterministic algorithms indeed never exceed the ϵ guarantee, and they usually obtain average error between $\frac{1}{4}\epsilon$ and $\frac{2}{3}\epsilon$. The maximum errors of Random and MRL99 are much smaller than ϵ , and the average errors are even smaller, revealing that their bounds are loose. We subsequently use the observed errors (max and average) as the primary error metric.

4.2.2 Space

Figure 6c, d shows the error-space trade-off of the algorithms using the max error and the average error, respectively. We see that MRL99 and Random are the best two algorithms with very similar performance. Between the two, MRL99

looks slightly better. This shows that the detailed choices of MRL99 offer a minor advantage, but not much. GKAdaptive and GKArray come quite close, especially when max error is considered. FastQDigest uses the largest space among all algorithms. Note that $\log u = 24$ in this case; we study other universe sizes subsequently.

4.2.3 Time

Figure 6e shows the trade-off between error and the update time per element for each algorithm. Here we use log scale on the x-axis, but linear scale on the y-axis. We see that for larger errors, all algorithms perform similarly, but GKAdaptive and FastQDigest degrades rapidly for smaller errors. This phenomenon can be better explained by the space-time trade-off plotted in Fig. 6f, in which we see that GKAdaptive and FastQDigest suffer a big speed loss when their space use exceeds 5 MB, which is roughly the size of CPU cache. This is because they perform a binary search for each incoming element, which is not cache-friendly. On the other hand, MRL99, Random, and GKArray still perform well, as they only use sorting and merging as their basic operations. Among these three, the two randomized algorithms are better than GKArray on larger errors, since this is where sampling kicks in. On smaller errors, the three algorithms have similar performance, with Random being slightly better than the other two.

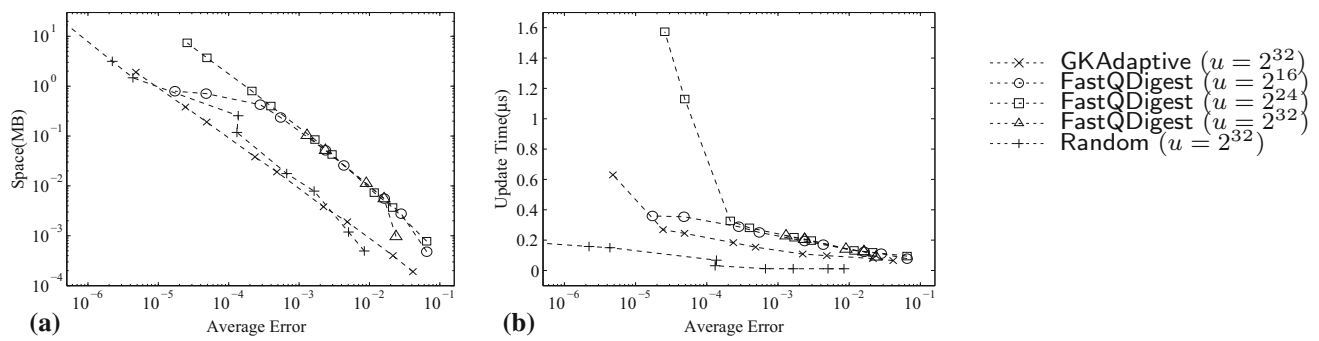


Fig. 7 Varying the universe size on normally distributed data, $\sigma = 0.15$. **a** Size-average error, **b** time-average error

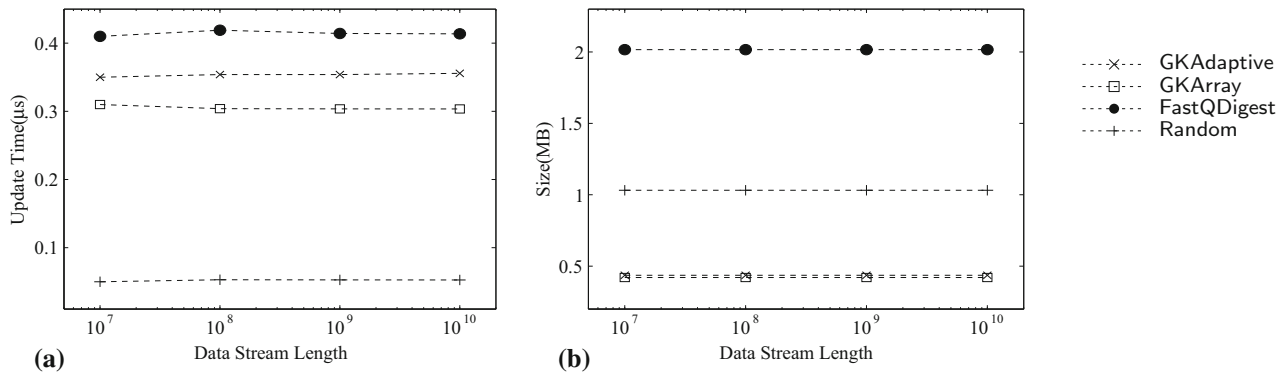


Fig. 8 Varying stream length on uniform distributed data, $u = 2^{32}$ and $\varepsilon = 0.0001$. **a** Time-stream length, **b** space-stream length

4.2.4 Varying universe size and data skewness

From the $O(\frac{1}{\varepsilon} \log u)$ bound, q-digest should work better with a smaller universe size. We tested the algorithms on synthetic data sets following a normal distribution, but with different universe sizes. The length of the stream is fixed at $n = 10^8$, and elements arrive in a random order. In Fig. 7a, b, we plot the error-space and error-time trade-offs of FastQDigest on data sets with different $\log u$. We also plot the curves of GKAdaptive and Random, the best deterministic and randomized comparison-based algorithms, which are unaffected by the universe size.⁵

From the figures, we see that q-digest is only competitive when $\log u = 16$ and $\varepsilon < 10^{-5}$. However, when this is the case, storing the frequencies of all the u elements exactly only takes 0.25 MB space. We also tested on data sets with different skewness by changing the variance of the normal distribution, but did not observe significant changes in the performance of q-digest. Therefore, we do not find any streaming situation where q-digest is the method of choice. Nevertheless, the algorithm remains of importance, since it is the only deterministic mergeable summary for quantiles [1], needed when summaries are merged in an arbitrary fashion.

⁵ It is possible for the error to be affected due to more duplicates in smaller universes, but we found this effect negligible in practice.

4.2.5 Varying stream length

We tested the algorithm on streams whose length increases from 10^7 to 10^{10} and plot how the time and space changes in Fig. 8a, b. We used uniformly distributed data, with the universe size fixed at $u = 2^{32}$ and $\varepsilon = 10^{-4}$. Elements arrive in a random order. We observe that there is little direct effect on update time or space usage as stream length grows, implying that these algorithms can scale to increasingly large data sets. Indeed, the per-element update time for Random actually *decreases*, due to random sampling playing a more major role as n goes up. The update time of the q-digest also goes down, since the cost of COMPRESS is amortized over more elements, as the algorithm only executes COMPRESS $\log n$ times throughout the whole stream.

Looking at Fig. 8b, we see that the space used by GKAdaptive and GKArray is essentially flat; we conjecture that they have a space bound independent of n on randomly ordered data. The space used by Random is constant, because the buffers are pre-allocated according to ε .

4.2.6 Sorted data

Finally, we tested how stream order affects the performance of the algorithms. We generated uniformly distributed data

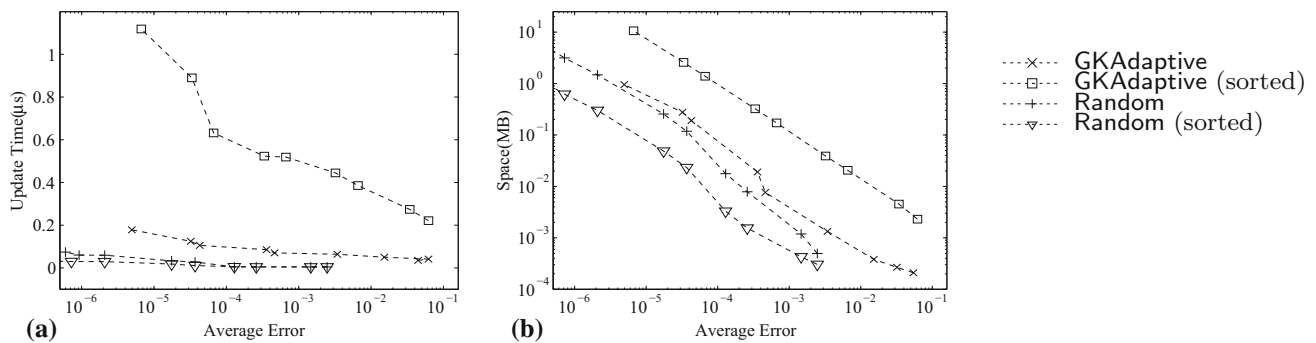


Fig. 9 Random order versus sorted—uniform distributed data, $u = 2^{32}$ and $n = 10^8$. **a** Time-average error, **b** space-average error

with $u = 2^{32}$ and $n = 10^8$ and compared performance on the sorted and randomly ordered streams. Here, we show the results for **GKAdaptive**, **Random**, the two most competitive algorithms.

In Fig. 9a, we see that when the stream order changes from random to sorted, the update time of **Random** decreases, while that of **GKAdaptive** increases, and the gap widens for smaller error. Recall that for **Random**, the amortized cost of all the merges per element is $O(1)$, while that of sorting is $O(\log(1/\varepsilon))$. When the stream is already sorted, the cost of the sorting goes down to $O(1)$ as well (the sorting algorithm we use in our implementation can make use of the existing sortedness in the data and reduce its work). On the other hand, when the stream is sorted, **GKAdaptive** has more trouble finding a removable element. When the stream is random ordered, very often the newly inserted element can be immediately removed. But when the stream is sorted, the heap always has to be checked, leading to a slower running time.

In terms of space, from Fig. 9b we see that **GKAdaptive** suffers from the sorted data because it has more trouble removing elements. However, **Random** is taking advantage of it, due to its way of sampling. When the data are sorted, the sampling part contributes almost no error at all because the sampled elements are equally spaced.

4.2.7 Conclusions for cash register algorithms

From our study, we can safely conclude that **GKArray** and **Random** are generally the best deterministic and randomized algorithm, respectively. **Random** is slightly better than **GKArray** in terms of both space and time, but the latter offers a worst-case guarantee on the error. However, note that we still lack a guarantee on its size as it uses a heuristic to remove tuples. On the other hand, **Random** uses a fixed amount of space that depends only on ε and should be used when there is a hard limit on space.

4.3 Results in the sliding window model

In this subsection, we evaluate the empirical performance of **Random** when it is plugged into the sliding window quantile algorithm of Arasu and Manku [3]. We denote by **AM** their original algorithm, which uses random sampling and the **GK** algorithm for each block (note that when ε is small enough, random sampling would not kick in). We used **GKArray**, which is the most competitive version of **GK** shown from our previous experimental results. We ran the experiments on the MPCAT-OBS data, set window size to be 1/2 of the full data set, and queried for all the quantiles when the window slides to the end of the stream.

From Fig. 10a, b, c, we observe similar results as in the full stream case, that is, **Random** achieves much better accuracy than **AM**, especially when average error is considered. This agrees with our analysis, which shows that **Random** does better in terms of the one-quantile guarantee, which essentially translates to the average error of the quantiles returned. The all-quantile guarantee is more closely related to the maximum error. Note that even though we cannot prove theoretically that **Random** is better than **AM** in terms of the all-quantile guarantee, the empirical results suggest that it is indeed better. This suggests that there is quite some looseness in the analysis.

Figure 10c, e plots the running time of the two algorithms as a function of the average error and space, respectively. This is also consistent with our earlier experimental results in the cash register model, which shows that **Random** is more efficient than **AM**. This is as expected since the sliding window algorithm is nothing but a series of instances of the cash register algorithm.

4.4 Results on turnstile algorithms

In this subsection, we compare the empirical performances of **DCM**, **DCS**, **Post** and **vPost**. We exclude the random subset sum sketch, as its performance is much worse than these algorithms.

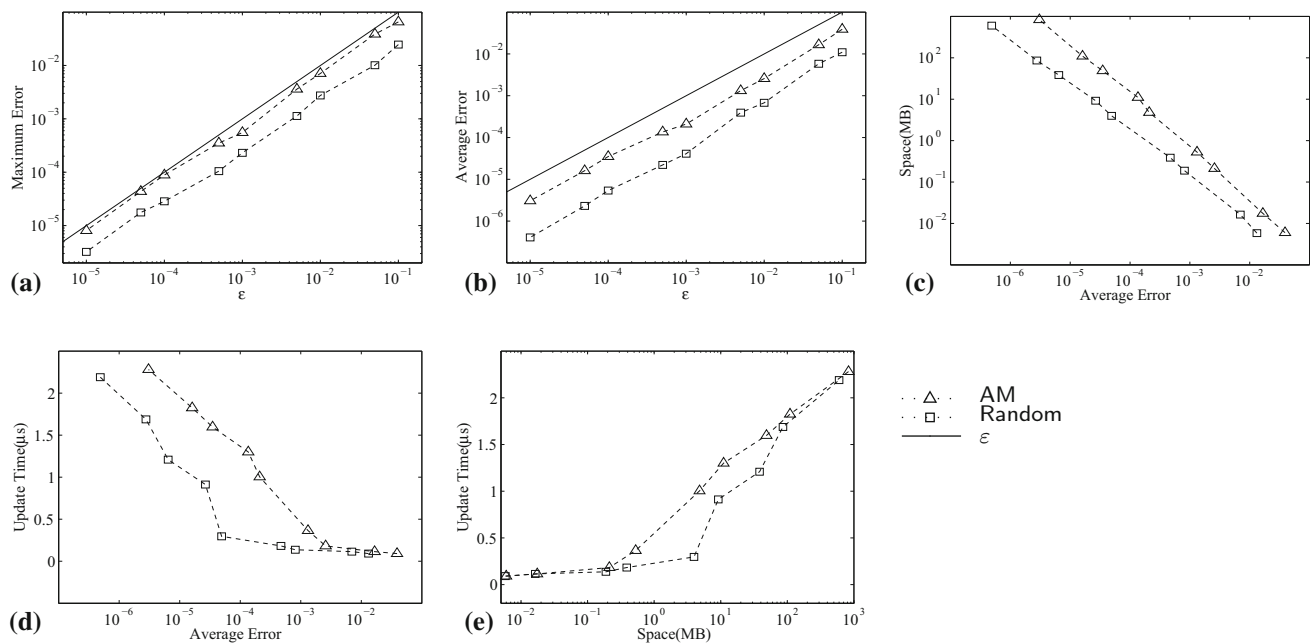


Fig. 10 Results on MPCAT-OBS for quantile algorithm over sliding window. **a** Maximum error- ϵ , **b** average error- ϵ , **c** space-average error, **d** time-average error, **e** time-space

Although we are experimenting with turnstile algorithms, it is not necessary to explicitly include deletions in the data sets: It is clear that the algorithms proceed in exactly the same way as on insertion-only data sets. Deleting a previously inserted element completely removes its impact on the data structure, so it has no effect on the accuracy, either. What matters is only those elements that remain. The only exception is *vPost*, whose behavior can be affected by deletions. However, as noted earlier, its performance can only be better when deletions are present, as they will make the invocation of the post-processing less frequent.

4.4.1 Parameter tuning

Recall that all the three algorithms use a sketch that is a $w \times d$ array, for each level in the dyadic hierarchy. Theoretically speaking, w determines the error, while d determines the confidence of obtaining an estimate within the error bound. In Sect. 3, we have given their relationships with the commonly used notion of an (ϵ, δ) -error guarantee. Intuitively, both w and d are meant to reduce the observed errors. So the question is, given a certain total sketch size, what is the best allocation to w and d ?

To this end, we first conduct a series of experiments trying out different combinations of w and d . Specifically, for a fixed sketch size, we vary d , which in turn determines w , and record the maximum and average errors of the computed quantiles. Here we used a uniformly distributed data set with $n = 10^7$ elements drawn from a universe of size $u = 2^{32}$. In Table 3,

Table 3 Tuning d for average error

d	Sketch size (KB)						
	64	128	256	512	1024	2048	4096
3	10.24	4.307	1.924	0.826	0.425	0.279	0.134
5	9.558	4.447	2.084	0.933	0.558	0.304	0.132
7	8.947	4.198	1.851	1.108	0.621	0.261	0.146
9	11.15	5.043	2.287	1.37	0.603	0.373	0.142
11	11.14	5.753	3.055	1.418	0.652	0.363	0.173
13	21.93	5.121	2.642	1.557	0.707	0.355	0.167

Bold sketch size is the minimum among each column

Table 4 Tuning d for maximum error

d	Sketch size (KB)						
	64	128	256	512	1024	2048	4096
3	53.67	22.92	9.27	7.71	3.58	2.56	0.931
5	50.04	25.11	11.13	8.07	3.383	2.498	0.931
7	65.26	22.28	8.71	5.49	2.923	1.693	2.419
9	75.41	27.39	8.87	9.543	2.63	2.389	0.542
11	61.03	33.32	13.5	8.769	3.067	2.261	0.824
13	139.3	29.25	17.34	7.503	2.843	1.824	0.869

Bold sketch size is the minimum among each column

we show the average errors ($\times 10^{-4}$) of DCS using a series of sketch sizes and find out that $d = 7$ appears to be a good choice. Similarly, we did the same for the maximum error in

Fig. 11 Trade-off between relative size and error. **a** Relative size– η , **b** relative error– η

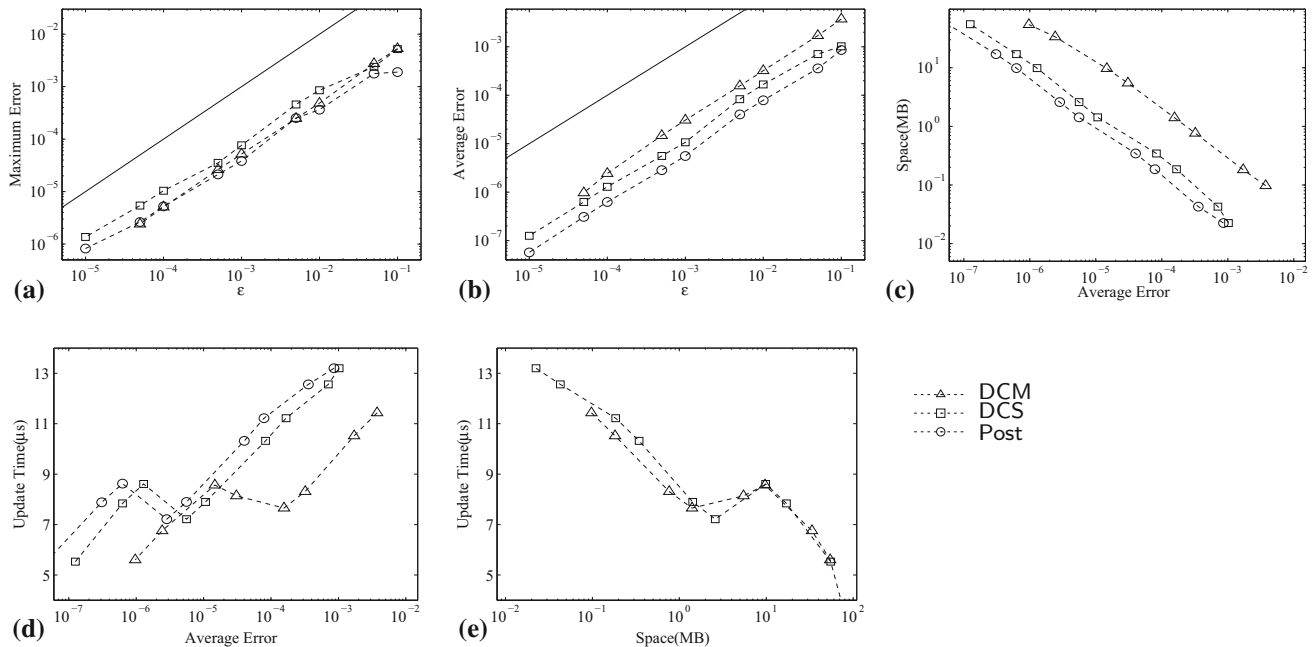
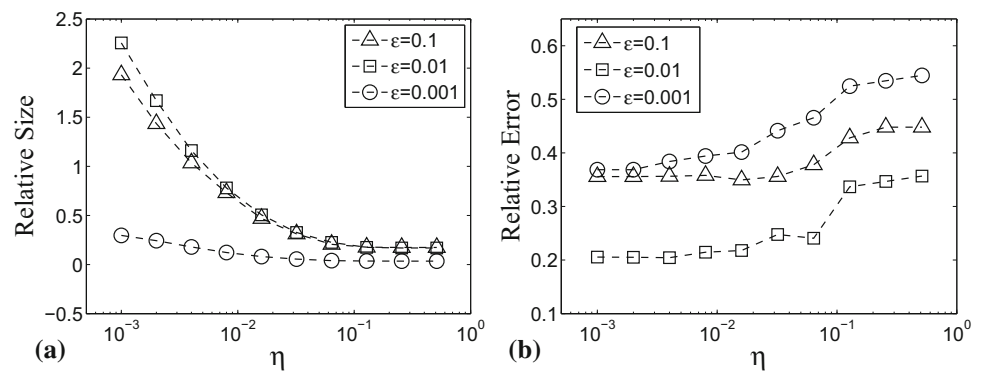


Fig. 12 Results on MPCAT-OBS. **a** Maximum error– ϵ , **b** average error– ϵ , **c** space–average error, **d** time–average error, **e** time–Space

Table 4. We observe that for the maximum error, we generally require a slightly larger d (which makes sense), but still 7 appears to be a good choice. We performed the same study for DCM and vDCS and found that $d = 7$ is the best choice there also. So we set $d = 7$ for all the subsequent experiments. We set $w = \frac{1}{\epsilon} \log u$ for DCM and $w = \sqrt{\log u}/\epsilon$ for DCS.

In Post, there is an additional parameter η which determines the trade-off between the size of the truncated tree $\hat{T}$ and the accuracy improvement. The tree size in turn determines the running time for the post-processing algorithm. We conducted a series of experiments by varying η on different values of $\epsilon = 0.1, 0.01, 0.001$ on our real data set. Figure 11 reports the size of $\hat{T}$ relative to that of the DCS sketch, as well as the reduced error relative to that of the original DCS sketch before the post-processing. From the results, we find that $\eta = 0.1$ is a sweet spot; further reducing η increases $|\hat{T}|$ without too much gain in terms of error reduction. We can see that our post-processing algorithm is quite effective,

reducing the error to 20–40 % of the original DCS sketch. It works better for larger ϵ , which also makes sense since DCS with a small ϵ is already quite accurate.

4.4.2 ϵ versus actual error

In Fig. 12a, b, we plot the actual maximum and average errors on the real data for different ϵ . This shows that the asymptotic analysis is rather loose: The actual maximum error is typically only $\epsilon/10$, while the average error is even smaller, and Post is quite effective at further reducing the error of DCS.

The actual errors of these three algorithms appear similar, but note that DCM has a larger size than DCS. Looking more closely at the curves, we see that DCM tends to be better in terms of the maximum error, but not as good in terms of average error. This might be due to the fact that the Count-Min sketch gives out biased estimators, while the

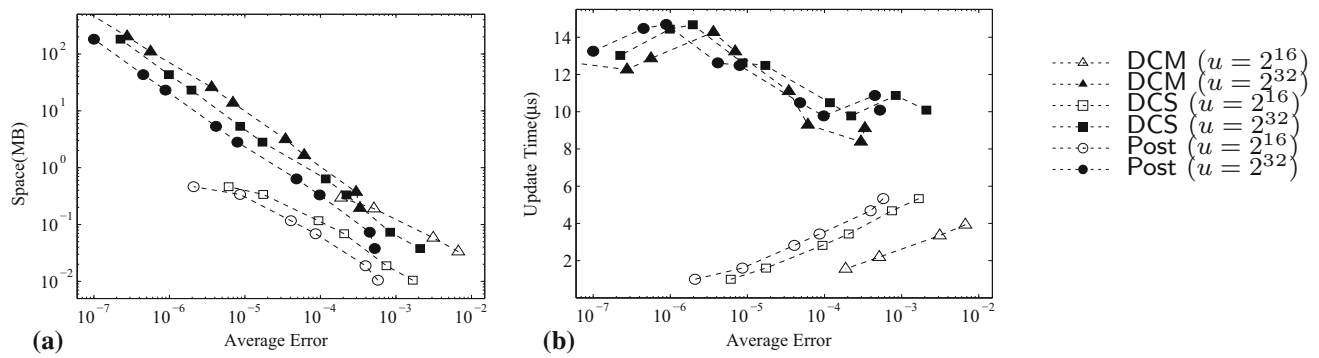


Fig. 13 Varying the universe size on the normal distributed data, $\sigma = 0.15$. **a** Space-average error, **b** time-average error

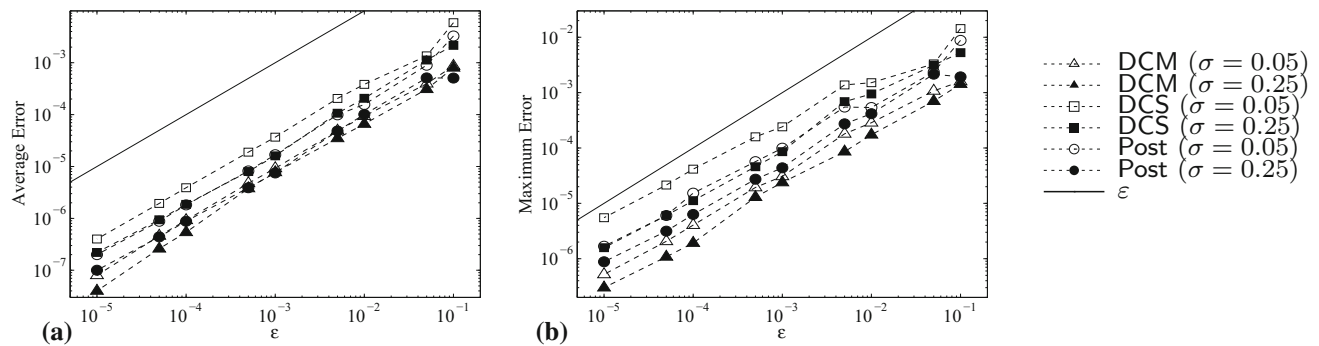


Fig. 14 Varying the deviation on the normal distributed data, $\sigma = 0.05, 0.25$. **a** ϵ -average error, **b** ϵ -maximum error

Count-Sketch is unbiased. Subsequently, we will use average error as the error metric unless specified otherwise.

4.4.3 Space

Figure 12c shows the error-space trade-offs of the algorithms. We see that to achieve the same error, DCS require only about 1/10 of the space required by DCM. While using the same amount of space, Post can further reduce the error by 60–80 %.

4.4.4 Time

Figure 12d shows the error-time trade-off. Note that since post-processing is only applied at the end of the stream and it is quite efficient, it has negligible impact on the amortized update time of DCS, so its curve is just that of DCS shifted to a smaller error.

In the space-time trade-off Fig. 12e, Post is thus identical to DCS, which is also very similar to DCM.

It is also instructive to compare Fig. 12c, d with Fig. 6c, e. This shows that the turnstile model is indeed more difficult to deal with than the cash register model. To achieve the same accuracy, the best turnstile algorithm has to spend significantly more space and time (roughly by an order of magnitude) than the best algorithm in the cash register model.

4.4.5 Varying universe size

The universe size u plays an important role in the turnstile algorithms, as it determines the height of the dyadic hierarchy. We tested the algorithms with data sets generated according to a normal distribution with $\sigma = 0.15$, but on different universe sizes. Figure 13a shows two series of trade-offs between error and space: One is on $u = 2^{16}$, and the other is on $u = 2^{32}$. Clearly, we see that a smaller universe indeed makes the algorithms more accurate, or equivalently speaking, makes the data structures smaller. The $u = 2^{16}$ curves halt at a small error value, since at this point the algorithms have sufficient space to store all frequencies exactly.

Similarly, Fig. 13b shows two series of trade-offs between error and update time for different universe sizes. Again, a small universe makes the algorithms much faster.

4.4.6 Varying data skewness

Finally, we tested the algorithms on data sets with different levels of skewness. We used data generated by a normal distribution with $\sigma = 0.05$ and 0.25 . Data skewness does not obviously affect space or time (for a given ϵ), so we only show how the actual errors respond, in Fig. 14a, b. From the figures, we see that as the data gets less skewed, the accuracy improves for all three algorithms. The improvement for DCM

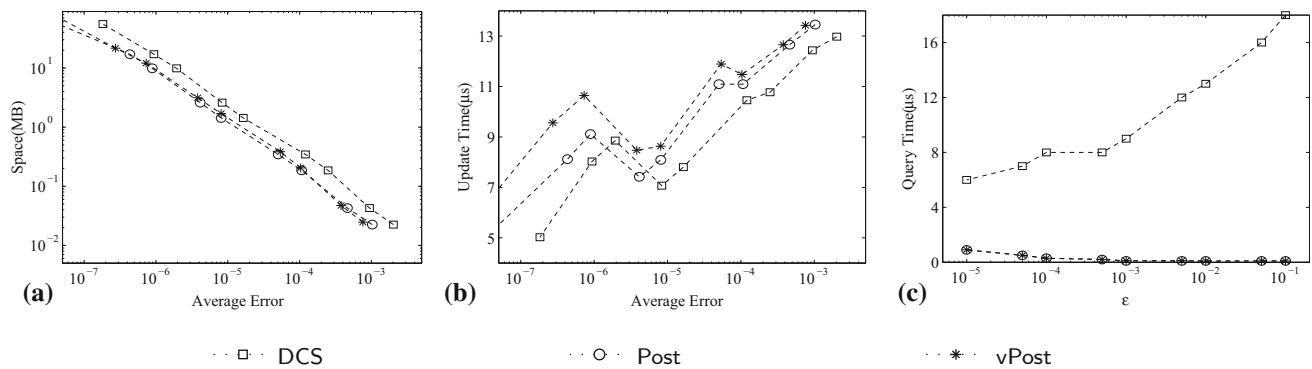


Fig. 15 Online version of post-processing results on MPCAT-OBS. **a** Space–average error, **b** update time–average error, **c** ϵ –query time

is very small, but it is more prominent for DCS and consequently for Post. This again is predicted well by the theory: Although in this paper, we analyzed the error of the Count-Sketch in terms of n in order to get the theoretical bound, its error actually depends more closely on the second frequency moment of the data, F_2 [5]. As the variance decreases, F_2 decreases, and the Count-Sketch gets more accurate. On the other hand, the Count-Min sketch does not depend directly on F_2 .

4.4.7 Post versus vPost

Finally, we examine empirical behavior of the online version of the post-processing method, denoted as vPost, in comparison with its offline version, Post, and the original DCS algorithm.

Recall that vPost invokes the post-processing algorithm periodically, whenever the current $\hat{T}$ no longer captures the change in the data. To make a fair comparison, we measured its accuracy by asking all-quantile queries at the middle point between the last two invocations. This should be a good indication of the typical performance of vPost. For Post, we performed the post-processing immediately before the queries were asked.

Figure 15a plots the trade-off between the space usage and the average error of the quantiles returned. By the parameters we have chosen, the size of the binary tree $\hat{T}$ is only about 1/10 of DCS. Thus, in a log scale, the space increase is barely visible. Interestingly, we observe that vPost has even smaller error than Post, albeit slightly. This is quite counterintuitive, as one would expect the accuracy to be the best right after the post-processing, and it will gradually degrade as new updates arrive. In fact, what really happens is that two opposite forces are at play: On one hand, new updates might increase the counters that are not kept in $\hat{T}$, which increases the error; on the other hand, for counters that are kept in $\hat{T}$, the new updates are *accurately* recorded, while they are only approximately recorded in the DCS sketch. However, it can be imagined that larger counters in the tree $\hat{T}$ play a

more important role than smaller counters. In our implementation, we invoke the post-processing whenever a leaf node's counter exceeds $2\eta\epsilon n = 0.2\epsilon n$, which is quite small. Thus, the benefit of having more accurate large counters overcome the accuracy loss due to not keeping the smaller counters. Put in another perspective, since we use 10 % more space to keep $\hat{T}$, it is actually understandable that we achieve a better accuracy. The net effect, therefore, is that we see two almost identical space–error trade-off curves in Fig. 15a.

In terms of update time, we do see a small (on average around 15 %) increase in vPost over Post, due to the cost of updating the counters in $\hat{T}$, as shown in Fig. 15b. Nevertheless, we feel that this increase in update cost can be justified by the 60–80 % reduction in error. Meanwhile, this increase in update time is necessary only when queries are to be supported at any time during the streaming process; when the queries arrive in batches, we can just apply the post-processing whenever needed, which removes the need to keep and maintain $\hat{T}$ at all times.

Finally, we also measured the query time of Post and vPost, in comparison with that of the original DCS sketch. From Fig. 15c, we see that the query time is much shorter after the post-processing. This is because after the post-processing, queries are answered from the binary tree $\hat{T}$, which is much smaller than the original DCS sketch, and avoids computing the hash functions in querying DCS sketch. This can be seen as an additional benefit of the post-processing method.

4.4.8 Conclusions for turnstile algorithms

From the experiments, it should be clear that DCS is the preferred turnstile algorithm for computing quantiles. DCM uses a much larger amount of space than DCS, while the running time of the two algorithms are similar. Finally, the post-processing algorithm is always beneficial to DCS, incurring no more space and time while being quite effective at further reducing the error. If queries are to be supported in an online fashion, vPost can be used, which achieves the

same space–error trade-off as **Post**, though the update time increases slightly.

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Additional lemmas and proofs

Size of the truncated tree $\hat{T}$

Lemma 1 *The truncated tree $\hat{T}$ has size $O(\frac{1}{\varepsilon} \log u)$ in expectation.*

Proof Recall that only nodes whose estimated frequency is above $\eta\varepsilon n$ are added to the truncated tree $\hat{T}$, where η is a constant. We classify these nodes into *heavy* nodes and *non-heavy* nodes. A node is heavy if its true frequency is above $\frac{1}{2}\eta\varepsilon n$, and non-heavy otherwise.

We first observe that, on any level of the dyadic structure, there are at most $\frac{n}{\frac{1}{2}\eta\varepsilon n} = O(1/\varepsilon)$ heavy nodes, treating η as a (small) constant. So even if they are all added to $\hat{T}$, there are only $O(\frac{1}{\varepsilon} \log u)$ of them in total.

For a non-heavy node v , the Count-Sketch may overestimate its frequency to above $\eta\varepsilon n$ with a constant probability, say $1/4$. Note that there can be $\Theta(u)$ non-heavy nodes being overestimated in expectation, but we will argue below that only $O(\frac{1}{\varepsilon} \log u)$ will be added during the top-down construction of $\hat{T}$.

Note that for a non-heavy node to be added, its parent must be a heavy node or another non-heavy node that has been overestimated. Thus, all the non-heavy nodes in $\hat{T}$ make up a number of subtrees, where the root of each subtree must be a child of a heavy node. Let $\mathbf{t}$ be any such non-heavy subtree, and we will bound $E[|\mathbf{t}|]$. Let r be the root of $\mathbf{t}$. For any node v below r , let I_v be an indicator variable where $I_v = 1$ if $v \in \mathbf{t}$ and $I_v = 0$ otherwise. Let $d(v)$ be the depth of v in $\mathbf{t}$, and we define $d(r) = 0$. For v to be added to $\mathbf{t}$, all of its $d(v)$ ancestors must have been added, so $E[I_v] \leq (1/4)^{d(v)+1}$ since each level uses an independent Count-Sketch. We then have

$$\begin{aligned} E[|\mathbf{t}|] &= \sum_{v \text{ below } r} E[I_v] \\ &= \sum_{v \text{ below } r} (1/4)^{d(v)+1} \\ &\leq \sum_{d=0}^{\log u} 2^d (1/4)^{d+1} \leq 1. \end{aligned}$$

Finally, we observe that at most two such $\mathbf{t}$'s can be attached to a heavy node, and there are only $O(\frac{1}{\varepsilon} \log u)$ heavy nodes, so we conclude that there are $O(\frac{1}{\varepsilon} \log u)$ non-heavy nodes in $\hat{T}$ in expectation. $\square$

Constraints on the x_i^* 's

Lemma 2 *Let x^* be the BLUE of (4). Let λ_v be any solution to (5). For any leaf w , let $Z_w = \lambda_w \sum_{z \in \text{anc}(w) \setminus r} y_z / \sigma_z^2$, and for any internal node v , $Z_v = \sum_{w < v} \lambda_w Z_w$. For any node v except the root r , let $F_v = \sum_{w \in \text{anc}(v) \setminus \{r\}} x_w^* / \sigma_w^2$. Let $\Delta = (Z_r - y_r \pi_r \text{'s left child}) / \lambda_r$. Then, we have*

$$\begin{cases} x_r^* = y_r, \\ x_v^* = (Z_v - \lambda_v F_{\text{parent}(v)} - \lambda_v \Delta) / \pi_v, \quad \text{for all nodes } v \neq r. \end{cases}$$

Proof We will follow the method of Lagrange multiplier [19] to find the BLUE of (4). Since only the subtree root is known exactly, we introduce a single Lagrange multiplier η . We set $\sigma_r^2 = 1/\eta$ instead of 0, and will later take the limit as η goes to ∞ . Denote by $\text{diag}(1/\sigma_v)$ the diagonal matrix with $1/\sigma_v$ at entry (v, v) . We further define $Z = \text{diag}(1/\sigma_v) y$ and $U = \text{diag}(1/\sigma_v) A$. Then, the Lagrange function can be rewritten as $(Z - Ux^*)^T (Z - Ux^*)$. By differentiation, we can derive (i) $y_r = x_r^*$; and (ii)

$$U^T U x^* = U^T Z. \quad (7)$$

This is sufficient to define a solution; we can solve for x^* by premultiplying by $(U^T U)^{-1}$, at the cost of a computing a matrix inverse. In the following, to derive the equations stated in the lemma from (7), which leads to a much more efficient algorithm to compute x^* .

Let $\text{anc}(u, v) = \text{anc}(u) \cap \text{anc}(v)$. Let u be any node of T_r . We also use $[\tau]$ denote the τ leaves of T_r . Then, by simple calculation, we can see that $(U^T U)_{u,w} = \sum_{v \in \text{anc}(u,w)} \sigma_v^{-2}$, and $(U^T Z)_u = \sum_{v \in \text{anc}(u)} y_v / \sigma_v^2$.

First, we take the weighted sum of corresponding rows on the LHS of (7) to obtain $\sum_{u < v} \lambda_u (U^T U)_{u,v} x^* =$

$$\begin{aligned} &\sum_{u < v} \sum_{z \in [\tau]} \left(\sum_{w \in \text{anc}(u,z) \setminus \text{anc}(v)} \frac{\lambda_u x_z^*}{\sigma_w^2} + \sum_{w \in \text{anc}(u,z) \cap \text{anc}(v)} \frac{\lambda_u x_z^*}{\sigma_w^2} \right) \\ &= \sum_{u < v} \sum_{z < w} \sum_{w \in \text{anc}(u,z) \setminus \text{anc}(v)} \frac{\lambda_u x_z^*}{\sigma_w^2} + \sum_{w \in \text{anc}(v)} \sum_{u < v} \sum_{z < w} \frac{\lambda_u x_z^*}{\sigma_w^2} \\ &= \sum_{u < v} \sum_{w \in \text{anc}(u) \setminus \text{anc}(v)} \frac{\lambda_u}{\sigma_w^2} \sum_{z < w} x_z^* + \sum_{w \in \text{anc}(v)} \sum_{u < v} \frac{x_w^*}{\sigma_w^2} \lambda_u \\ &= \sum_{u < v} \sum_{w \in \text{anc}(u) \setminus \text{anc}(v)} \lambda_u x_w^* / \sigma_w^2 + \lambda_v \sum_{w \in \text{anc}(v)} x_w^* / \sigma_w^2. \end{aligned} \quad (8)$$

Note that in the last line of (8), the second component can be written as

$$\lambda_v \sum_{w \in \text{anc}(v)} \frac{x_w^*}{\sigma_w^2} = \lambda_v \left(F_{\text{parent}(v)} + \frac{x_v^*}{\sigma_v^2} + \frac{x_r^*}{\sigma_r^2} \right).$$

We can also derive that the first component is

$$\sum_{u < v} \sum_{w \in \text{anc}(u) \setminus \text{anc}(v)} \frac{\lambda_u x_w^*}{\sigma_w^2} = \left(\pi_v - \frac{\lambda_v}{\sigma_v^2} \right) x_v^*.$$

To see this, let us assume that this holds for any descendant of v . Then, we can derive $\sum_{u < v} \sum_{w \in \text{anc}(u) \setminus \text{anc}(v)} \frac{\lambda_u x_w^*}{\sigma_w^2}$

$$\begin{aligned} &= \sum_{\{s \text{ is a child of } v\}} \sum_{u < s} \left(\frac{\lambda_u x_s^*}{\sigma_s^2} + \sum_{w \in \text{anc}(u) \setminus \text{anc}(s)} \frac{\lambda_u x_w^*}{\sigma_w^2} \right) \\ &= \sum_{\{s \text{ is a child of } v\}} \left(\frac{\lambda_s x_s^*}{\sigma_s^2} + \sum_{u < s} \sum_{w \in \text{anc}(u) \setminus \text{anc}(s)} \frac{\lambda_u x_w^*}{\sigma_w^2} \right) \\ &= \sum_{\{s \text{ is a child of } v\}} \left(\frac{\lambda_s x_s^*}{\sigma_s^2} + \left(\pi_s - \frac{\lambda_s}{\sigma_s^2} \right) x_s^* \right) \\ &= \sum_{\{s \text{ is a child of } v\}} \pi_s x_s^* = \pi_v x_v^* = \left(\pi_v - \frac{\lambda_v}{\sigma_v^2} \right) x_v^*. \end{aligned}$$

Combining the above two results, we have

$$\sum_{u < v} \lambda_u \left(U^T U \right)_u x^* = \pi_v x_v^* + \lambda_v \left(\frac{x_r^*}{\sigma_r^2} + F_{\text{parent}(v)} \right). \quad (9)$$

Secondly, we take the weighted sum of corresponding rows on the RHS of (7) to obtain

$$\begin{aligned} \sum_{u < v} \lambda_u \left(U^T Z \right)_u &= \sum_{u < v} \lambda_u \sum_{w \in \text{anc}(u)} \frac{y_w}{\sigma_w^2} \\ &= \sum_{u < v} \lambda_u \left(Z_u + \frac{y_r}{\sigma_r^2} \right) \\ &= Z_v + \frac{\lambda_v y_r}{\sigma_r^2}. \end{aligned} \quad (10)$$

Finally, by combining (9) and (10), we derive that $\forall v$,

$$\pi_v x_v^* = Z_v - \lambda_v F_{\text{parent}(v)} - \lambda_v (x_r^* - y_r) \eta. \quad (11)$$

Substituting v by r in (11), we can derive

$$x_r^* = \frac{\left(\frac{Z_r}{\eta} + y_r \lambda_r \right)}{\left(\frac{\pi_r}{\eta} + \lambda_r \right)}.$$

We already have $x_r^* = y_r$. Then, we can conclude that either $\eta = +\infty$ or $y_r \pi_r = Z_r$. As we know, y_r is given and irrelevant to $\pi_r Z_r$ which implies $\eta = +\infty$. To handle this infinity, we first express

$$\Delta(\eta) = (x_r^* - y_r) \eta = (Z_r - y_r \pi_r) (\pi_r / \eta + \lambda_r),$$

where s is a child of root r . Now we take limit of $\Delta(\eta)$ and derive that

$$\Delta = \lim_{\eta \rightarrow +\infty} \Delta(\eta) = (Z_r - y_r \pi_r) / \lambda_r.$$

Finally by taking limit on (11) for any $v \neq r$, we derive $x_v^* = (Z_v - \lambda_v F_{\text{parent}(v)} - \lambda_v \Delta) / \pi_v$, and the lemma is proved. $\square$

Covariance analysis

Lemma 3 Suppose we build a Count-Sketch with $d = 1$ row and w columns on a vector x . For any two elements $u \neq v$, let y_u and y_v be the estimators for x_u and x_v . Then, $\text{Cov}(y_u, y_v) = x_u x_v / w$.

Proof Recall that in the Count-Sketch, for any element v , the estimator is computed as $y_v = g(v) \sum_z g(z) x_z I_v(z)$, where $I_v(z) = 1$ if $h(v) = h(z)$ and 0 otherwise. So we have

$$\begin{aligned} \text{Cov}(y_u, y_v) &= E(y_u y_v) - E(y_u) E(y_v) \\ &= E \left[\left(g(u) \sum_{h(i)=u} g(i) x_i \right) \left(g(v) \sum_{h(j)=v} g(j) x_j \right) \right] \\ &\quad - x_u x_v \\ &= \sum_{i,j} E[g(u) g(v) g(i) g(j)] x_i x_j E[I_u(i) I_v(j)] - x_u x_v \end{aligned}$$

Let $f(i, j) = E[g(u) g(v) g(i) g(j)] x_i x_j E[I_u(i) I_v(j)]$. We find that $f(u, v) = x_u x_v$ and $f(v, u) = x_u x_v / w$ (note, the last term in $f(i, j)$ is not symmetric in i and j). If $\{i, j\} \neq \{u, v\}$, then $f(i, j) = 0$, since $g(\cdot)$ is 4-wise independent hash function. Therefore, we derive that $\text{Cov}(y_u, y_v) = x_u x_v / w$. $\square$

On the other hand, prior analysis on the Count-Sketch shows that $\text{Var}(y_v) = \frac{1}{w} \sum_i x_i^2$. Thus, the covariance is usually order of magnitude smaller than the variance.

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